



# Guardians of the Genome: p53 and the SMC5/6 Complex

*Rebekah Napier-Jameson*

*PostDoc Wylie Lab*

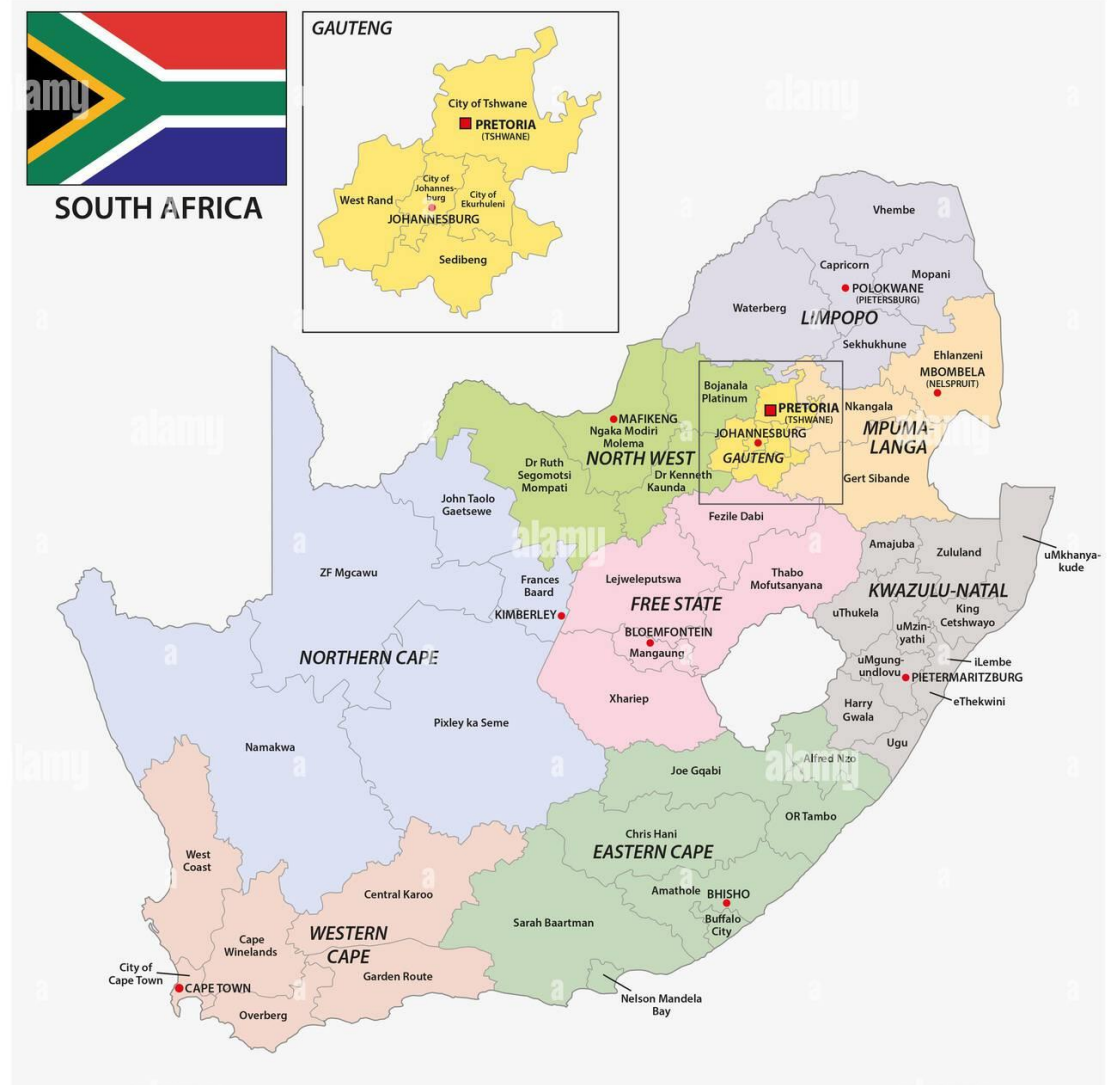
*SMU Department of Biological Sciences*

*07/15/2026*

*QuarkNet Workshop*

# Who am I?

- I grew up in South Africa





# Who am I?

- I wanted to go into Horticultural Sciences after high school



# Who am I?

- I came over to the **USA** on a swimming scholarship



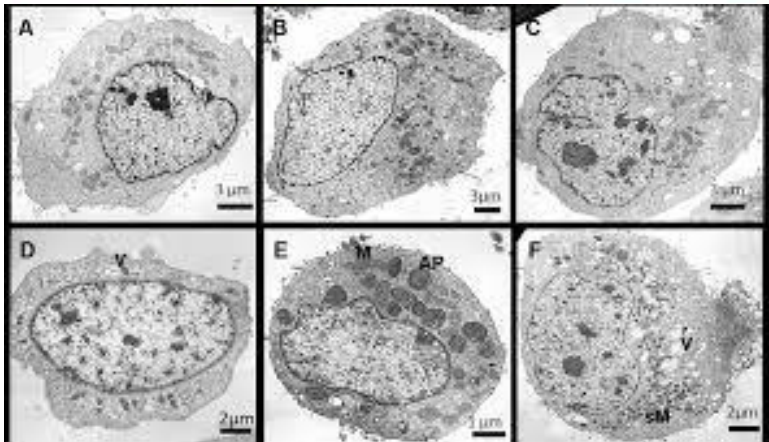
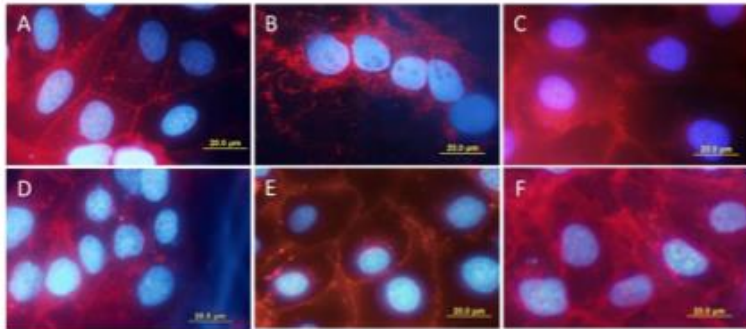
# Who am I?

- **BSc Biology (pre-med, with Chemistry minor) and BSc Environmental Sciences (Wildlife Management)**
  - Student worker (Chemical inventory, greenhouse caretaker, lab set up and tear-down, microbiology research assistant, teaching assistant and worker at Center for Interdisciplinary Geospatial Information Technologies)





# Who am I?



- **MS Biotechnology**

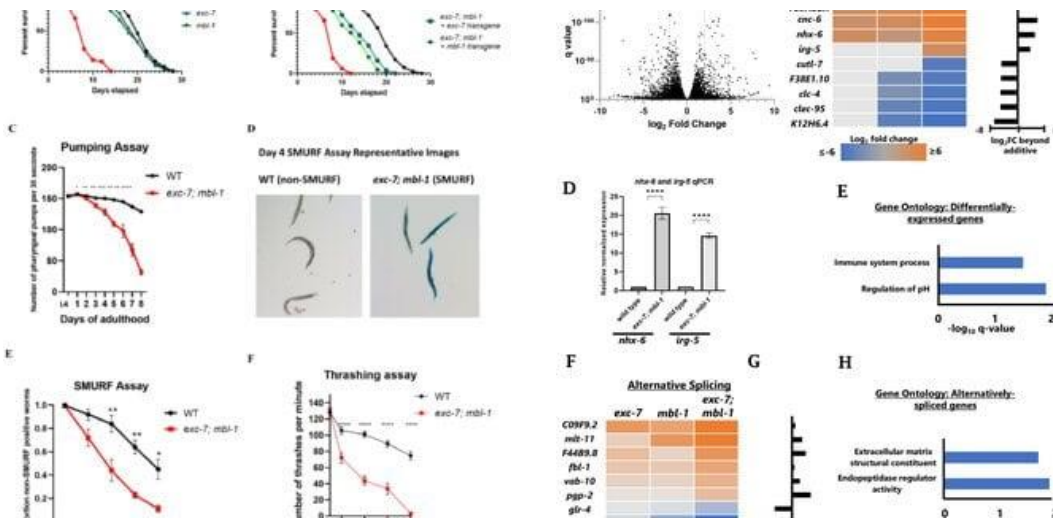
- The comparison of Effects of Synthetic and Natural Arachidin-3 on Rotavirus Infected Cells
- Research assistant and TA (Summer- Pre-nursing and Concepts of Human Biology)

# Who am I?

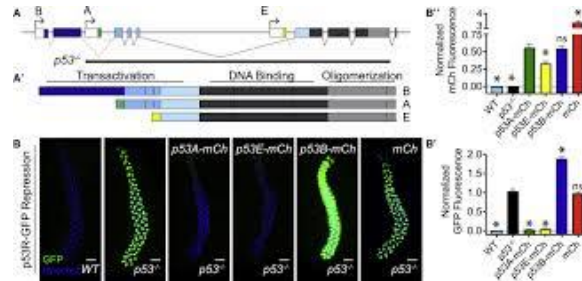


- **PhD Molecular and Cellular Biology**

- Regulation of mRNA and circRNA as a Cause and Consequence of Aging in *Caenorhabditis elegans*
- Research assistant and TA (Introductory Biology Lab I & II)



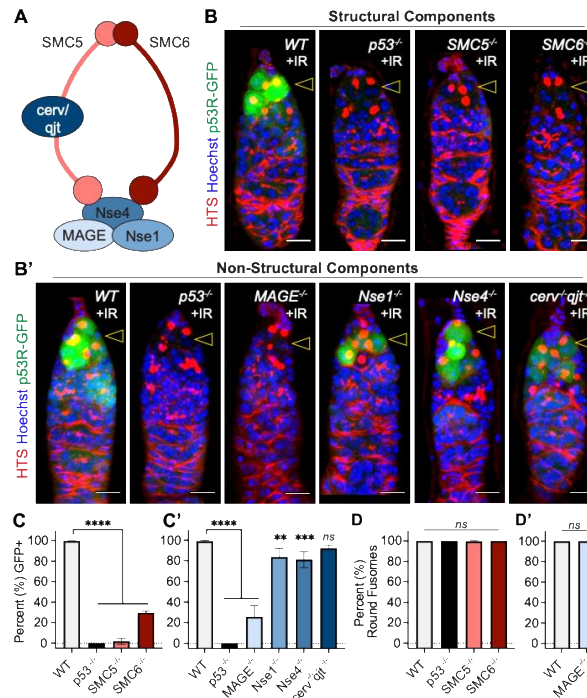
# Who am I?



## • Post Doctoral Fellow in the Wylie Lab at SMU

- Engineering novel triple-tagged p53 Drosophila Strain to enable isoform-specific functional analysis.

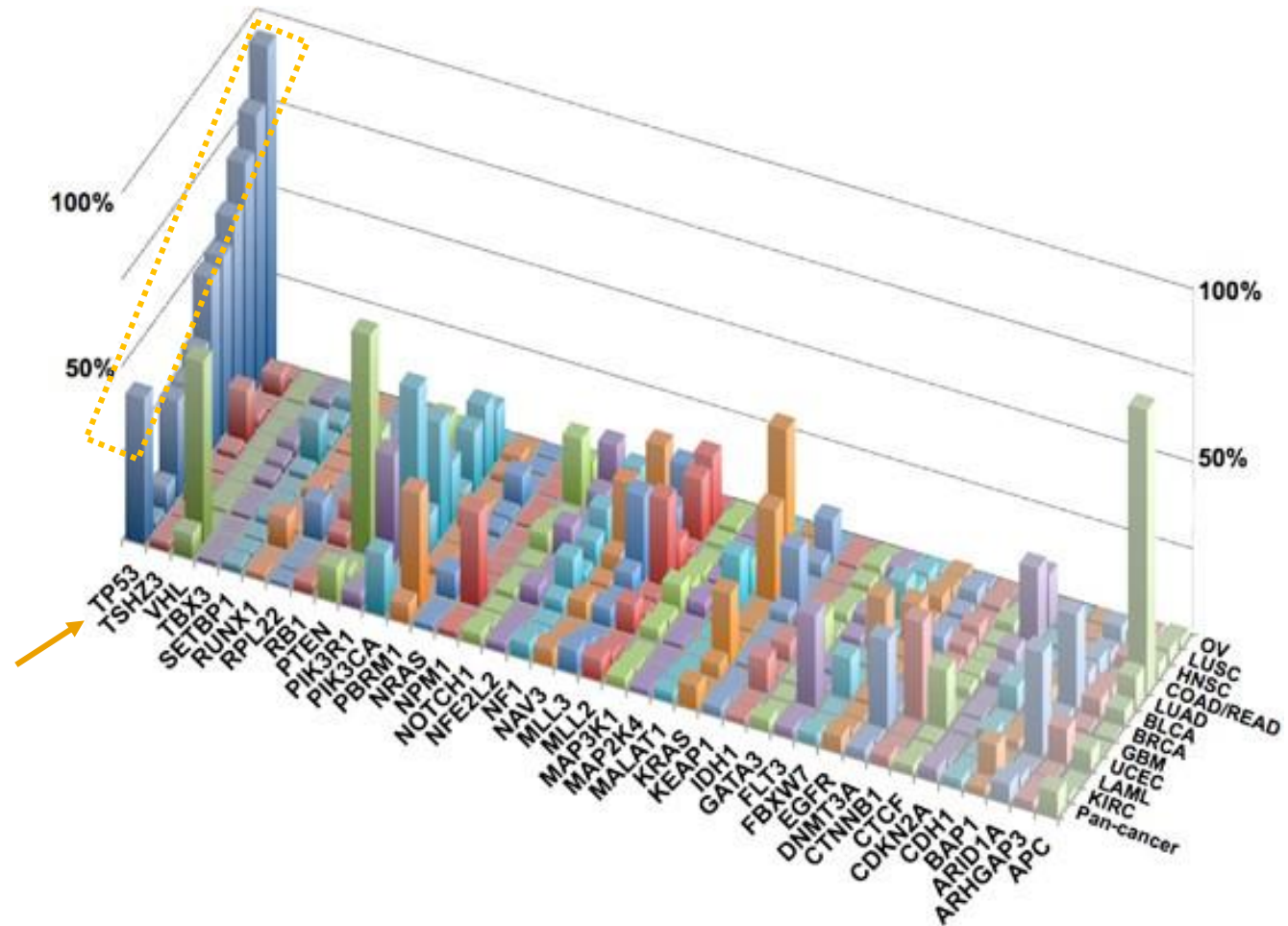
- Investigating the role of the SMC5/6 complex in regulating p53 activity, uncovering new mechanisms of stem cell genome stability



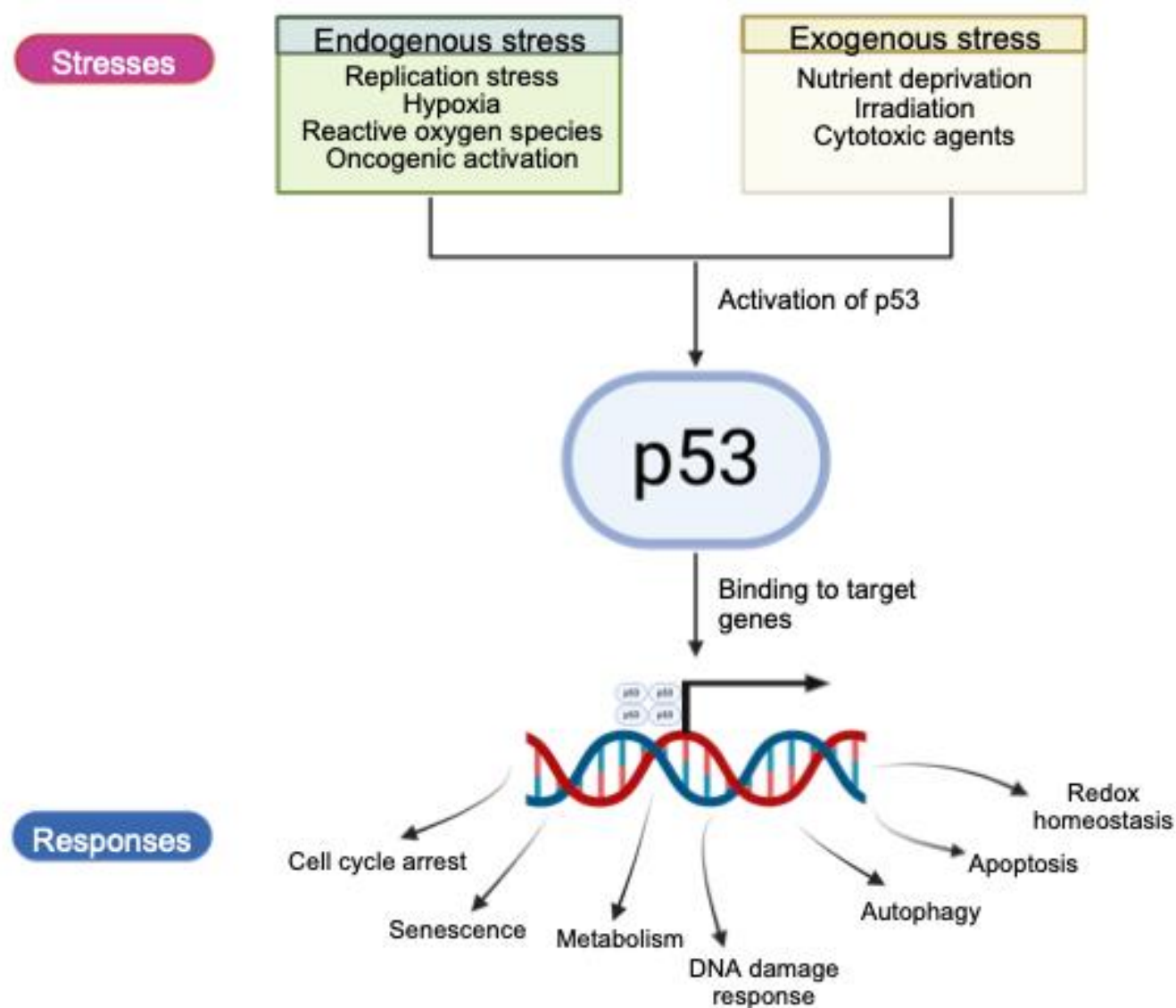


# What is p53?

- Most commonly mutated gene in human cancers. One of the most well-studied genes in the human genome, having been cited in more than 10,000 papers since its discovery in 1979
- Despite this there is still much unknown about p53

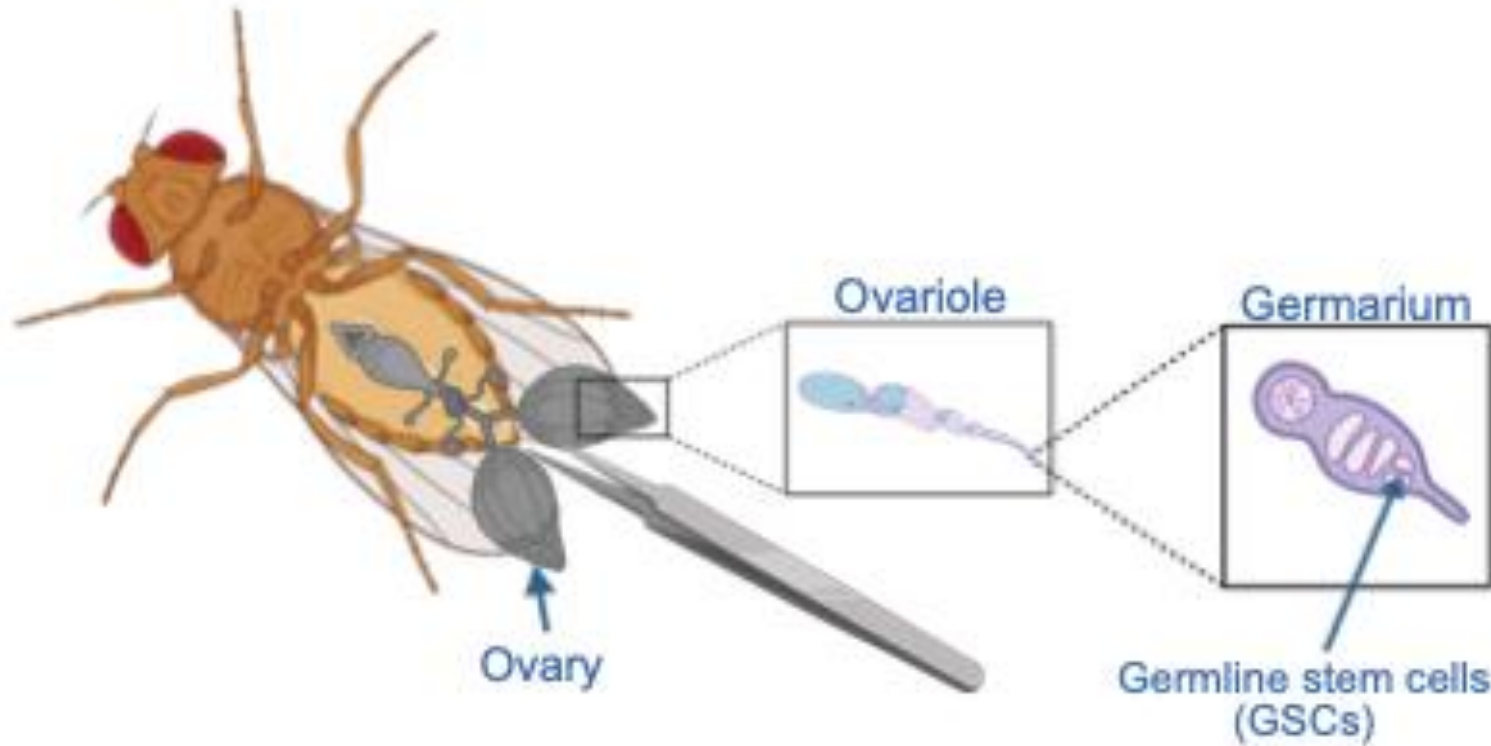


# What does p53 do?



# How does the Wylie Lab study p53

*Drosophila melanogaster*



Egg lay cage

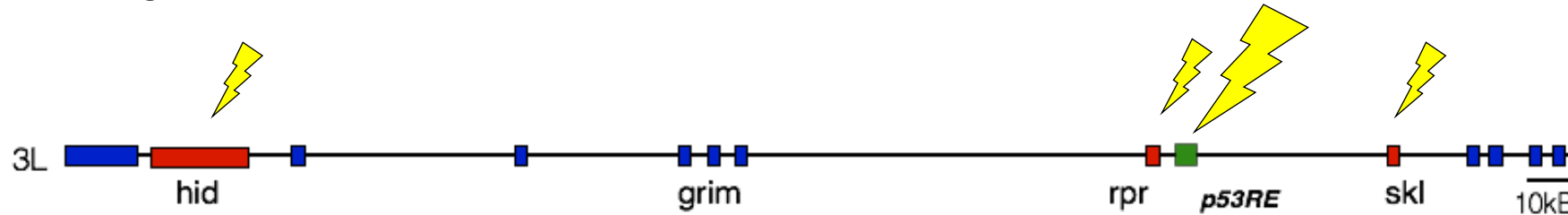


+ cell culture and mouse work (not shown today)

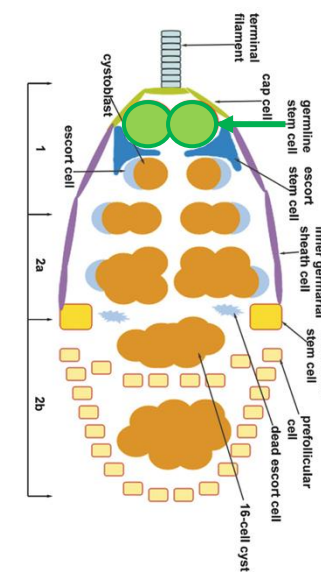
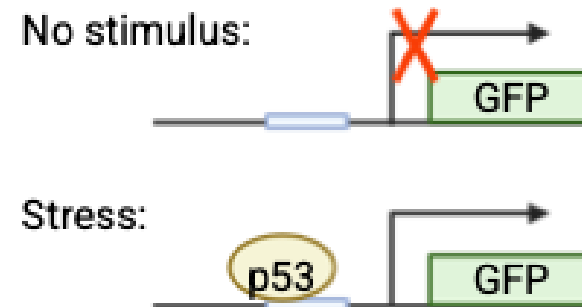
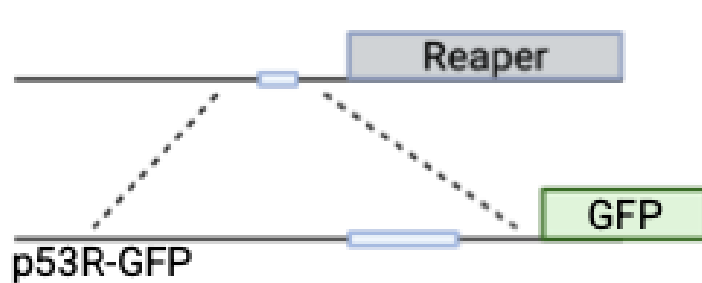


# How does the Wylie Lab study p53

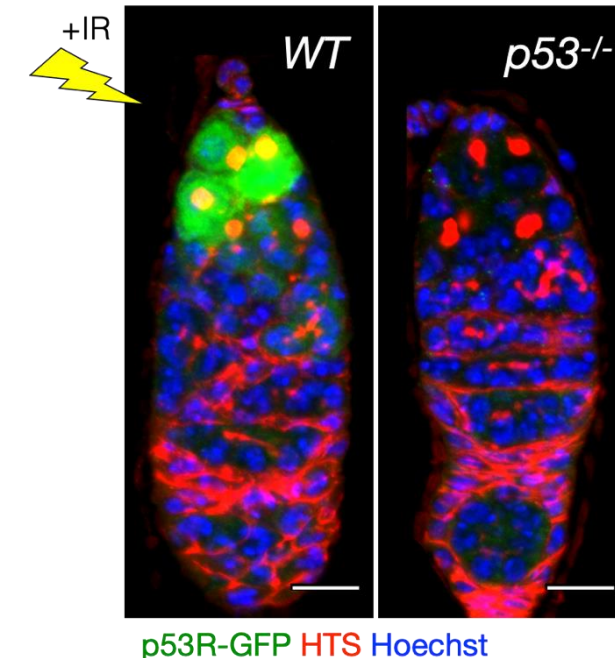
cell death gene locus:



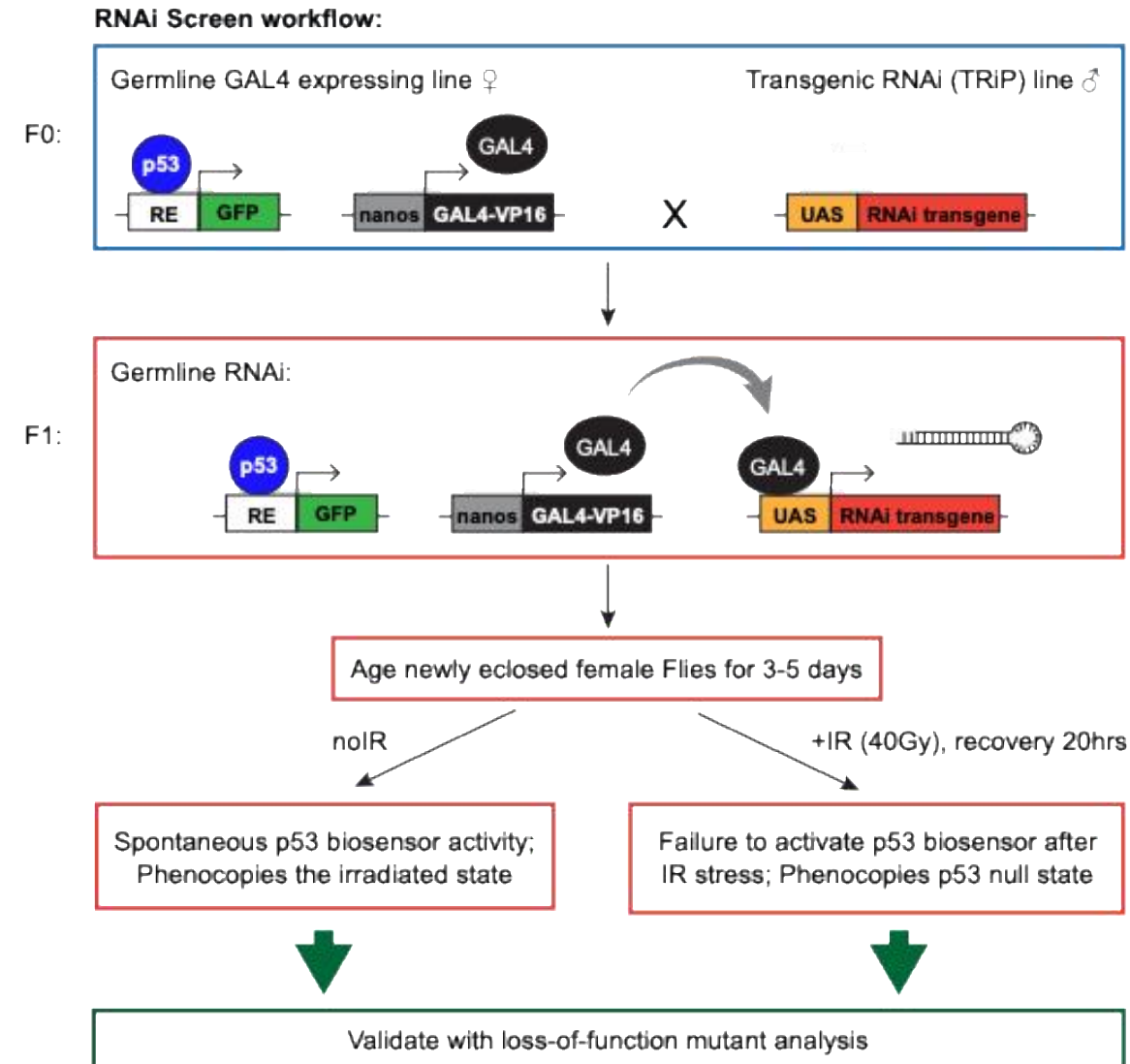
*p53 cell killing mediated by a single enhancer*



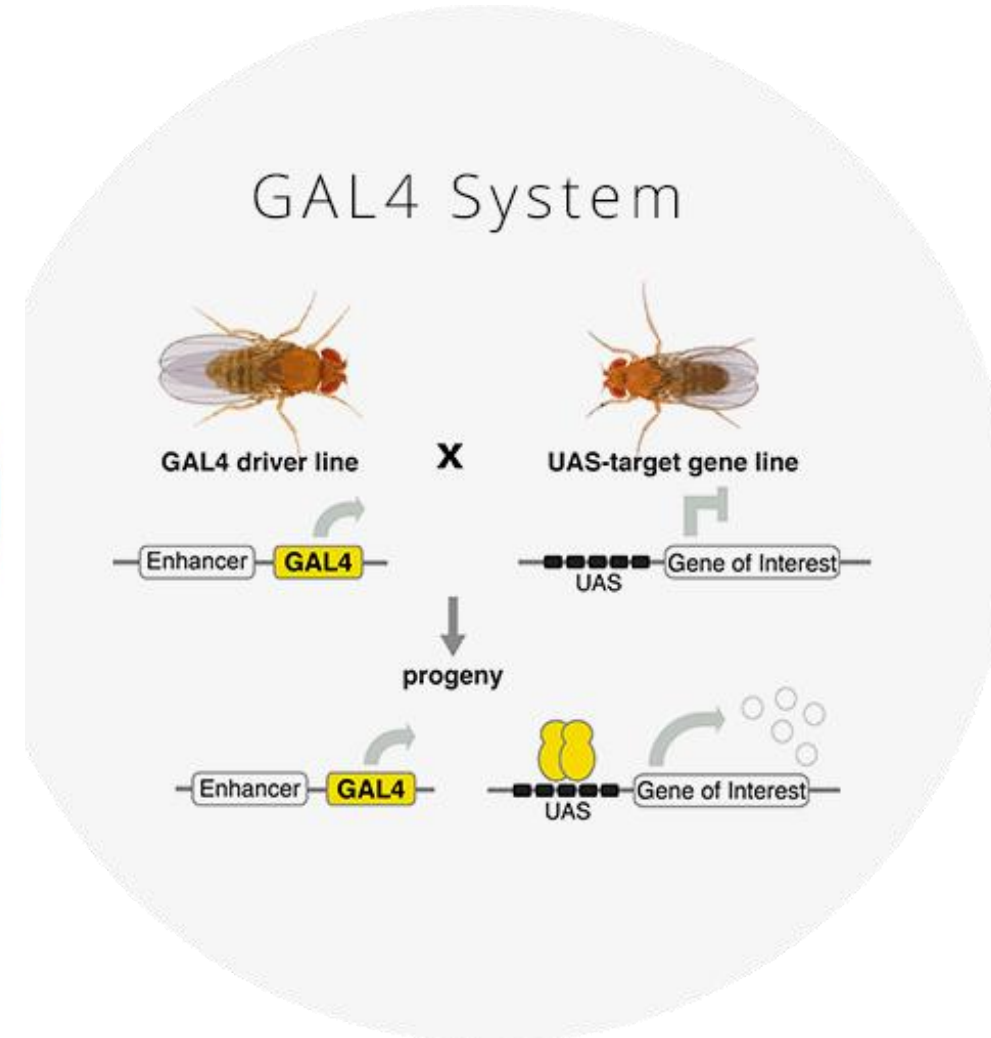
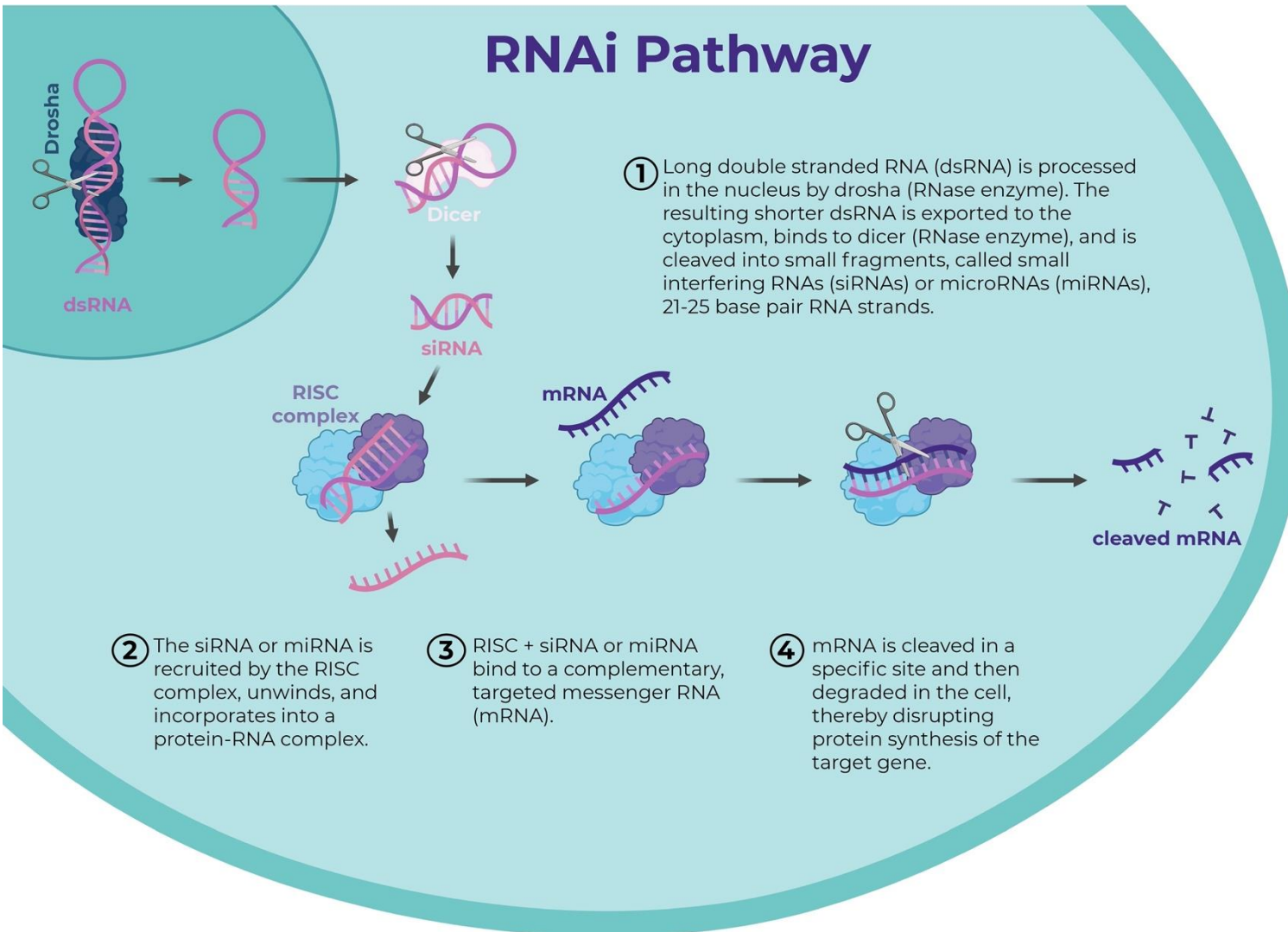
Adapted from: Wu, Xiaodong et al.  
"Drosophila follicle cells: morphogenesis in an eggshell." *Seminars in cell & developmental biology* 19 3 (2008): 271-82.



# Are there other genes that phenocopy p53's failure to activate?

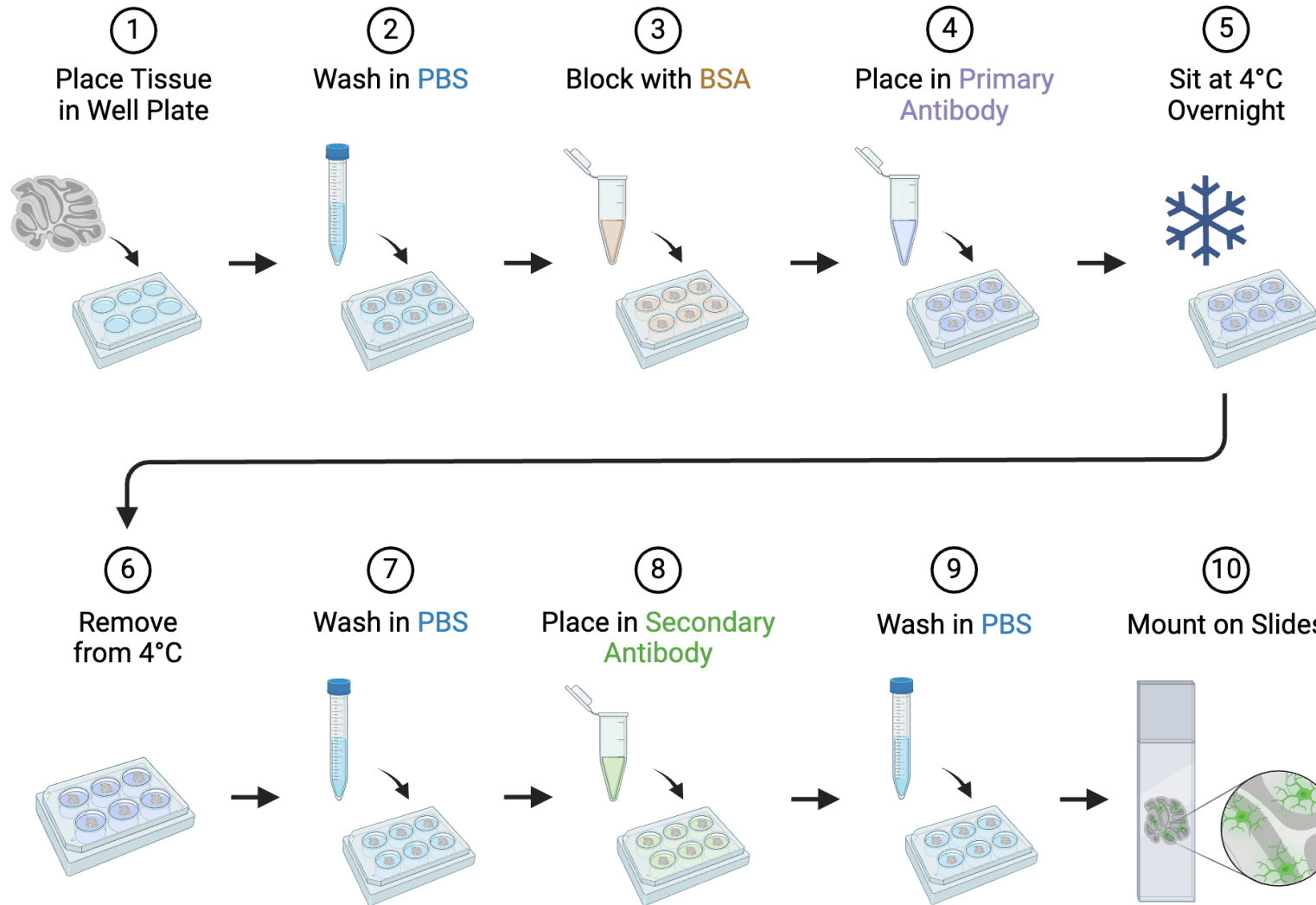


# What tools are needed for our screen?





# What is Immunohistochemistry (IHC)?

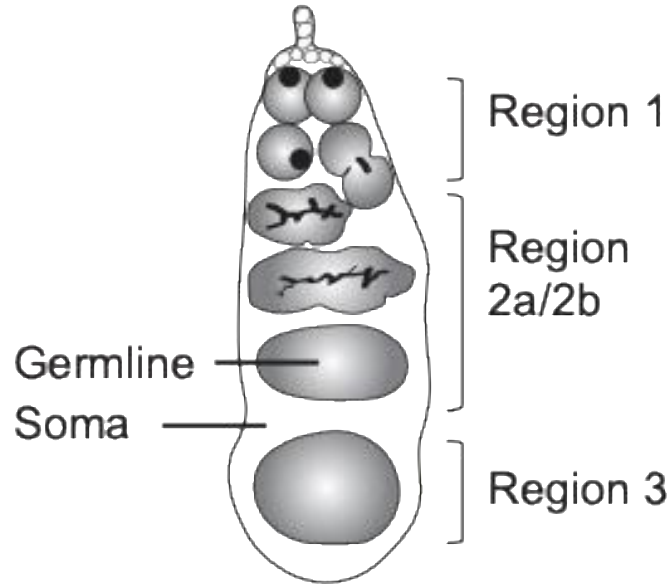


Indirect



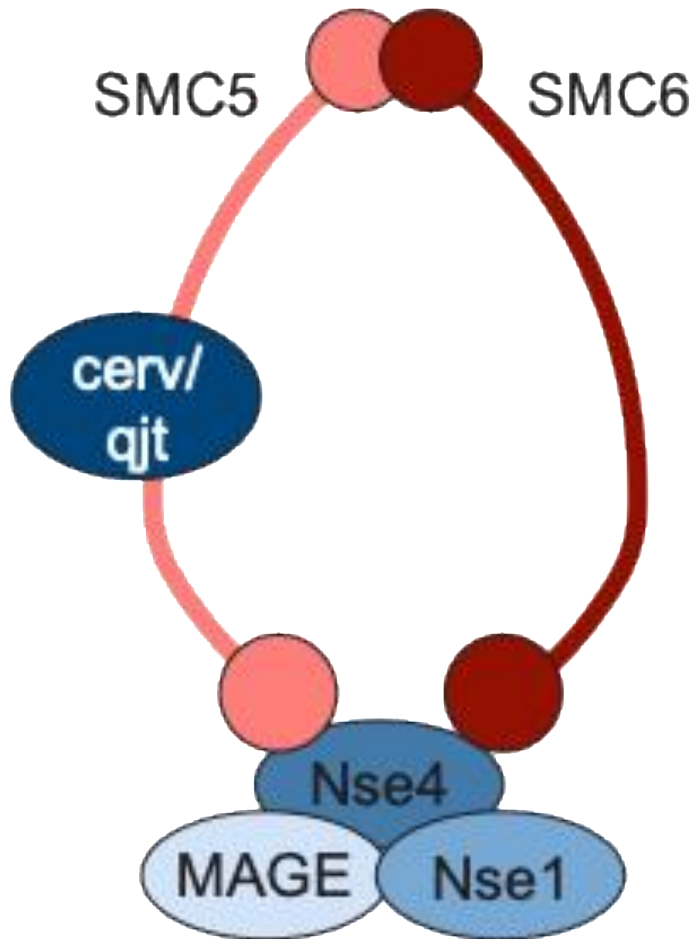
- Reporter enzyme or fluorochrome
- Y Labeled Primary Antibody
- Y Primary Antibody
- Antigen

# What candidates did the screen find?



SMC6 phenocopies p53 in terms of the  
“Failure to activate p53R-GFP after IR stress” phenotype!

# What is the SMC5/6 complex



- Maintains genome stability during DNA replication, repair, and chromosome segregation
- Core heterodimer: SMC5 + SMC6
- Non-SMC5/6 components: Some have been shown to have SUMO E3 ligase or ubiquitin ligase activity
- Broadly conserved
- Mutations associated with many disease states, including cancer
- Link to p53 activity not well established

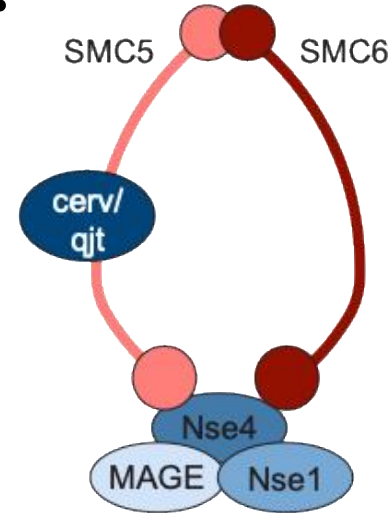
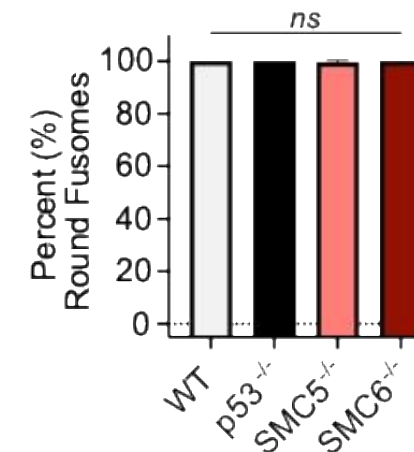
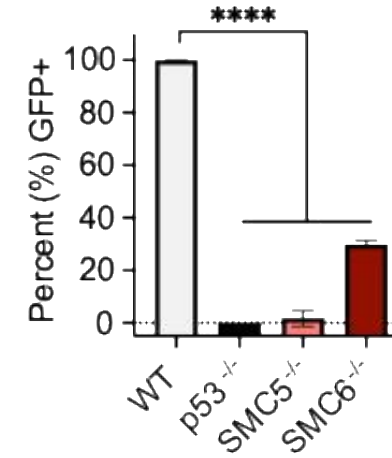
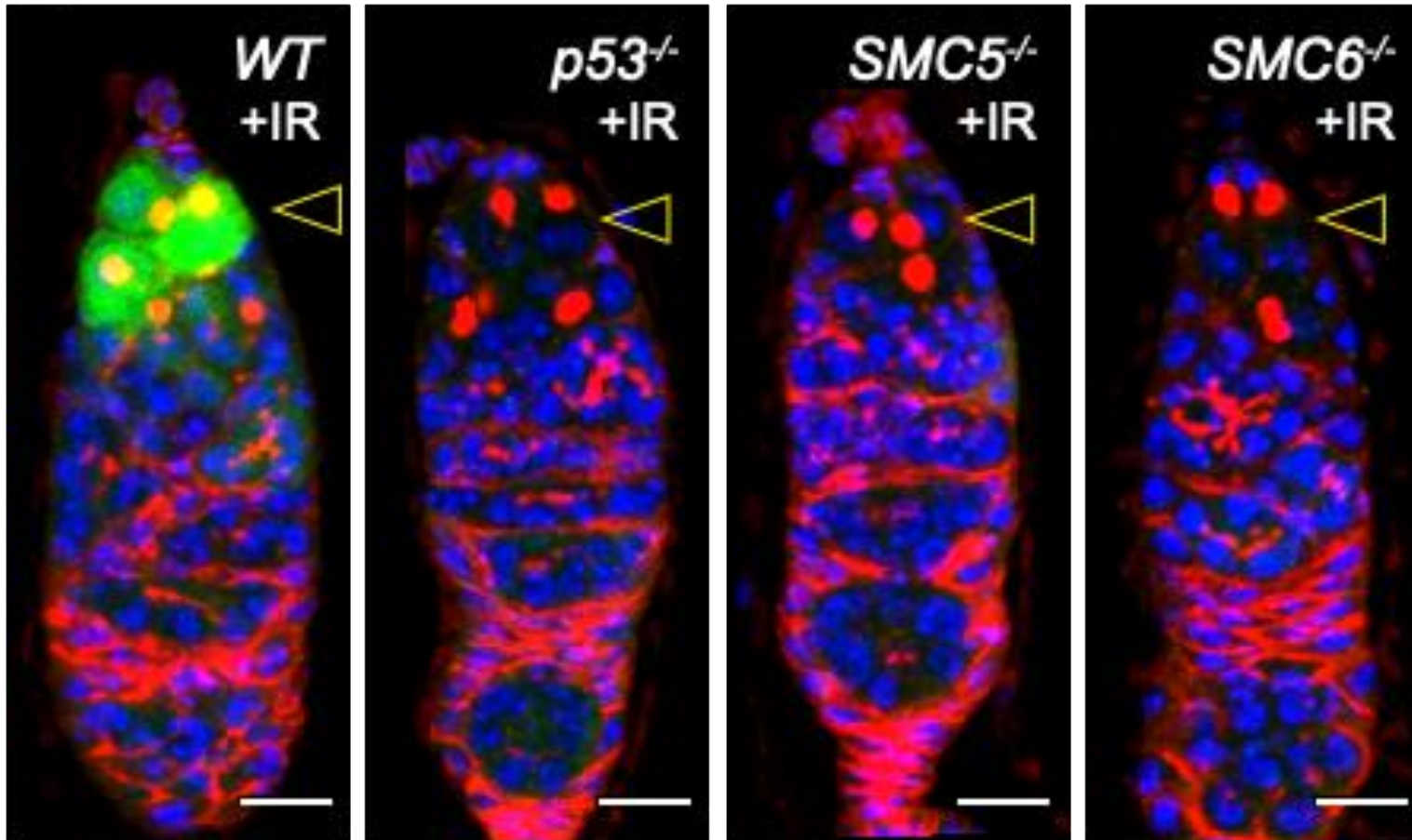
Conserved complex that safeguards genome integrity by coordinating DNA replication, repair, recombination, and chromosome organization, particularly under conditions of replication stress and DNA damage



# Do both SMC5/6 components phenocopy p53 loss?

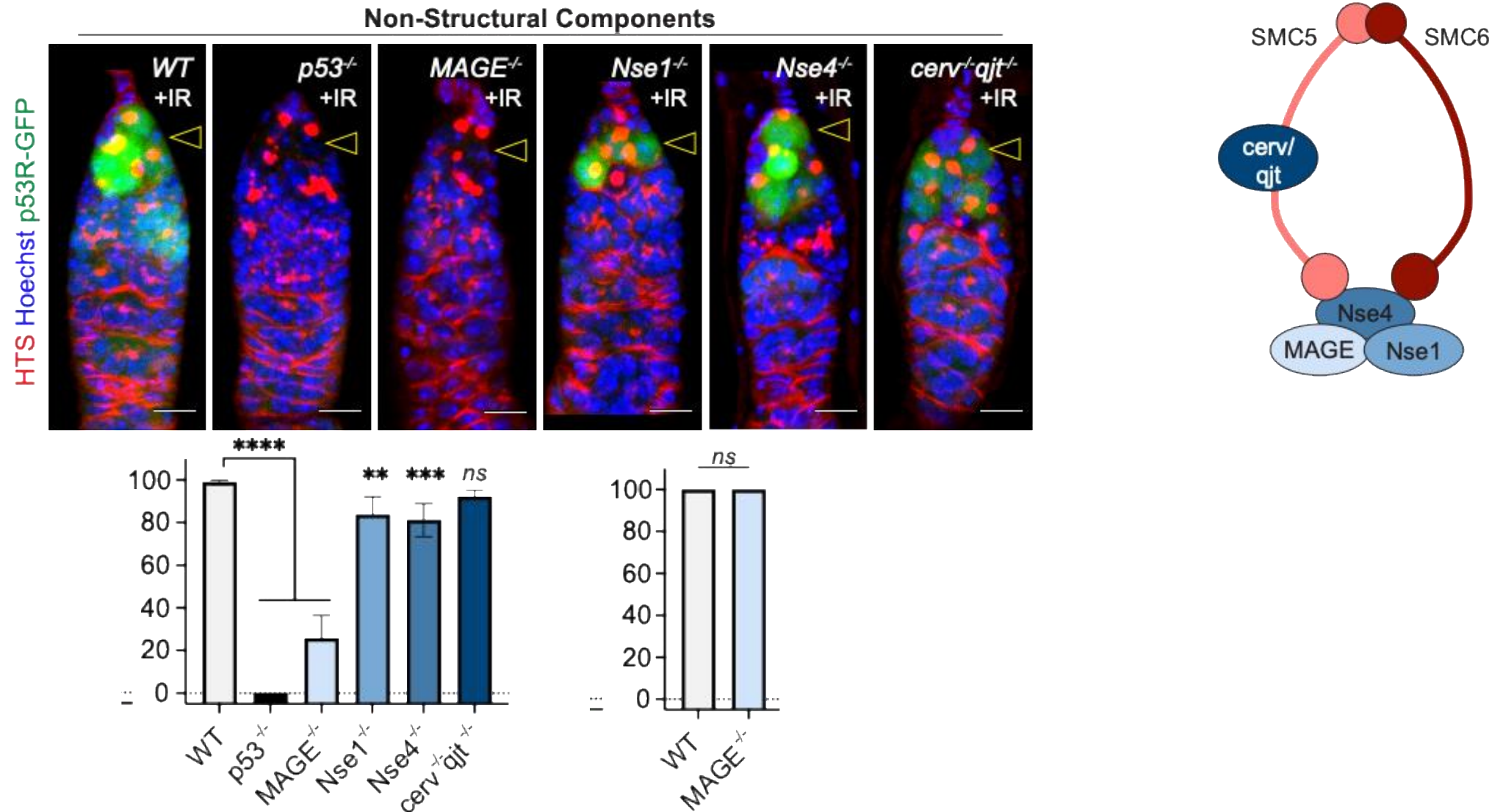
## Structural Components

HTS Hoechst p53R-GFP



Through mutant analysis we show that p53 activity in the germline stem cell compartment requires the SMC5/6 complex

# Do the non-SMC5/6 components phenocopy p53 loss?

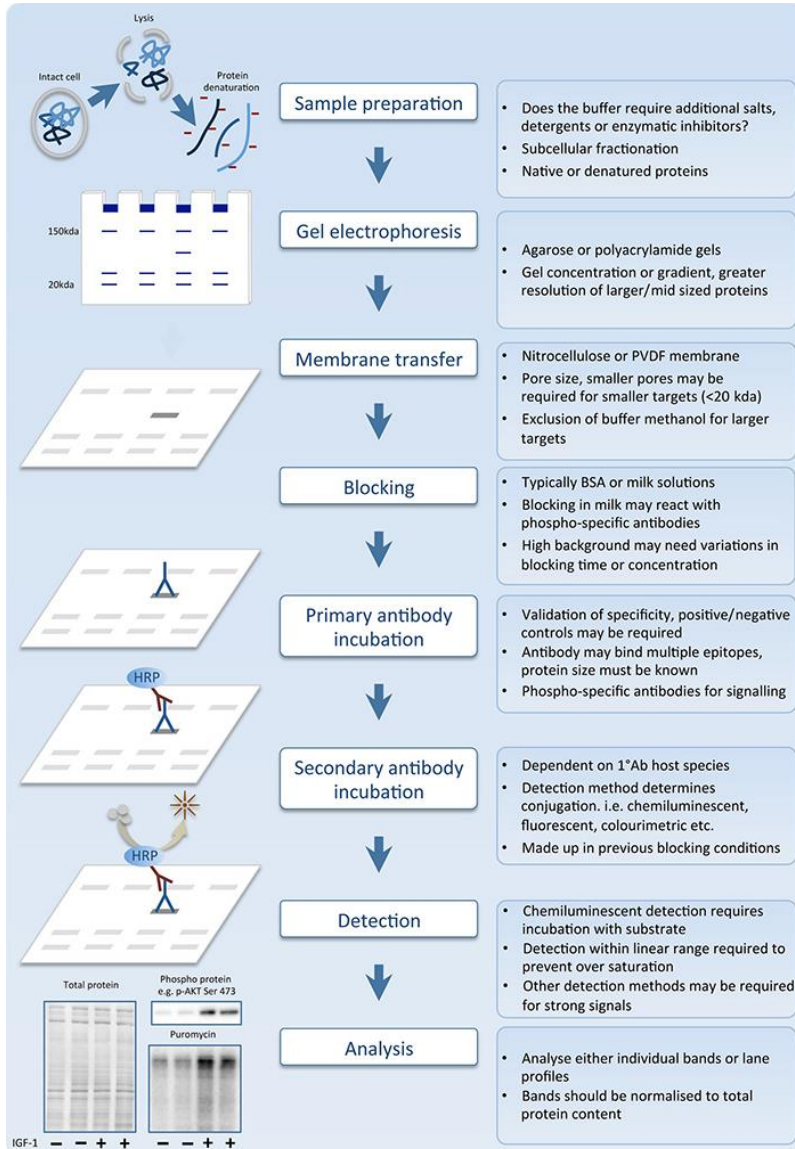


The non-SMC5/6 components have varying effects on the “Failure to activate p53R-GFP after IR stress” phenotype

# How can be probe for proteins?

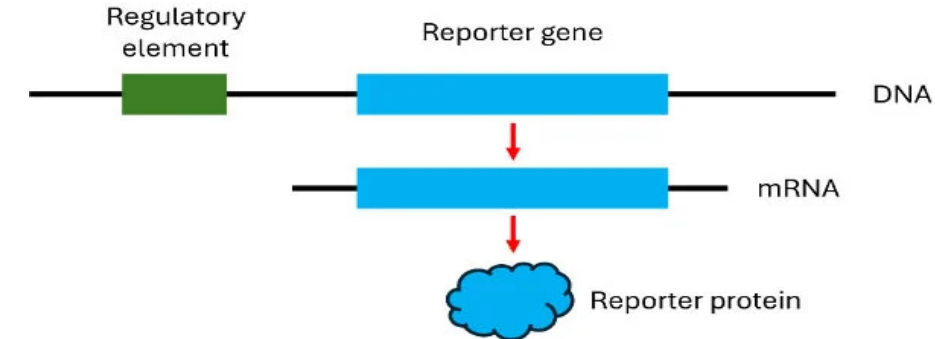


## Western Blotting



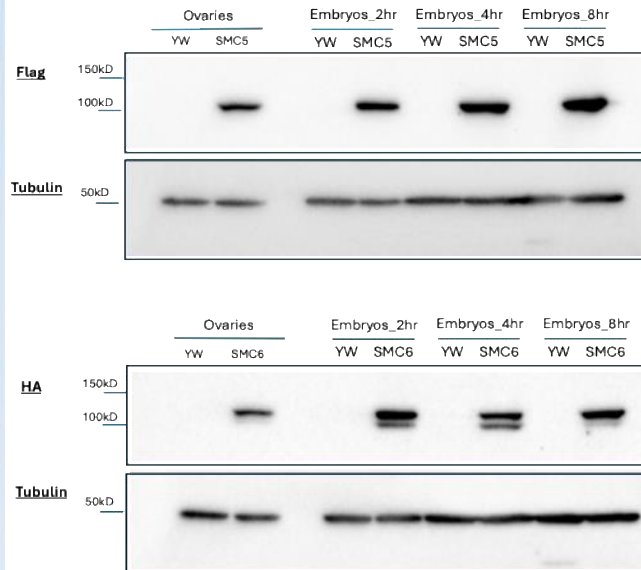
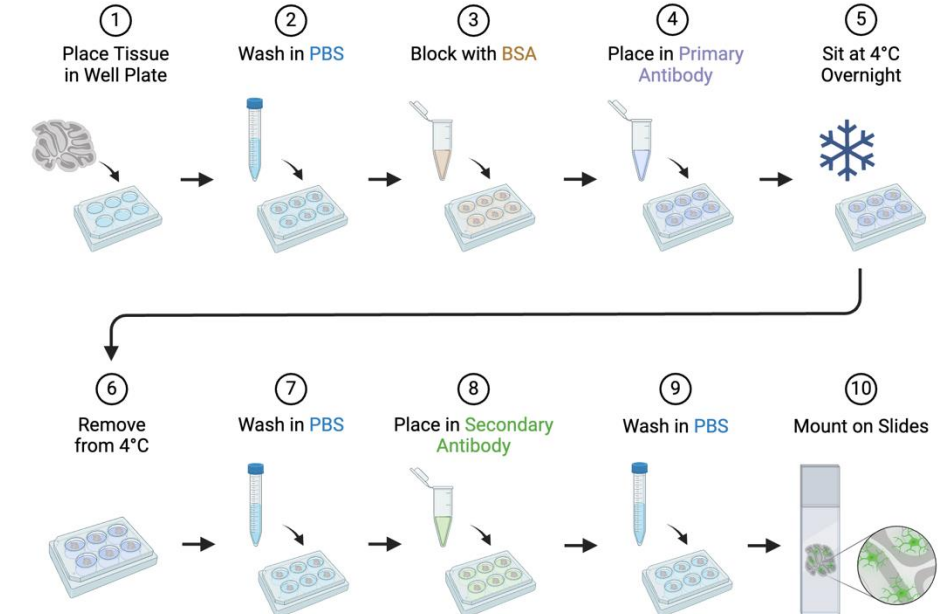
<https://precisionbiosystems.com/western-blot-troubleshooting-guide/>

## Reporters



<https://www.biocompare.com/Editorial-Articles/624233-Introduction-to-Reporter-Cell-Lines/>

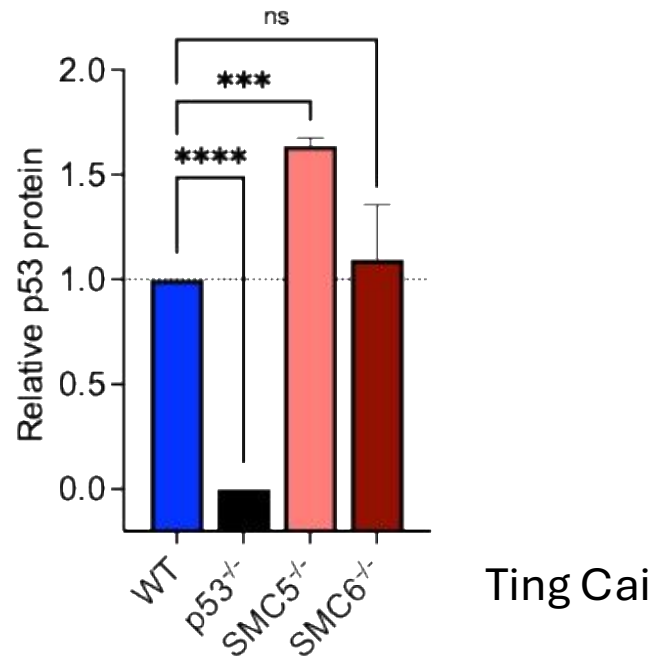
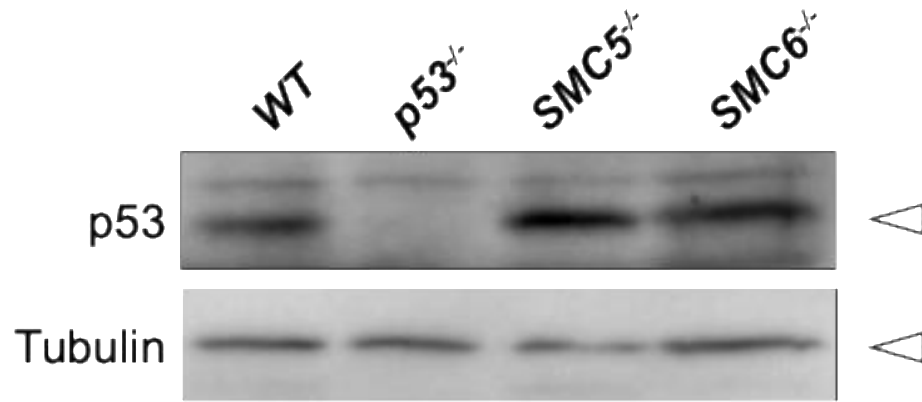
## IHC



Napier-Jameson R, Cai T, Johnson A, Wylie A. The SMC5/6 complex specifies p53 activity in the *Drosophila* germline stem compartment. *Under revision for submission.*



# Is p53 present in SMC5 and SMC6 mutants? SMU | DEDMAN COLLEGE OF HUMANITIES & SCIENCES

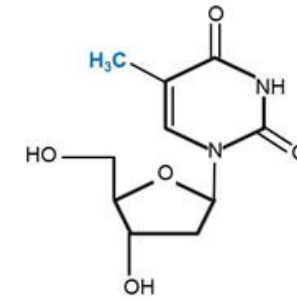
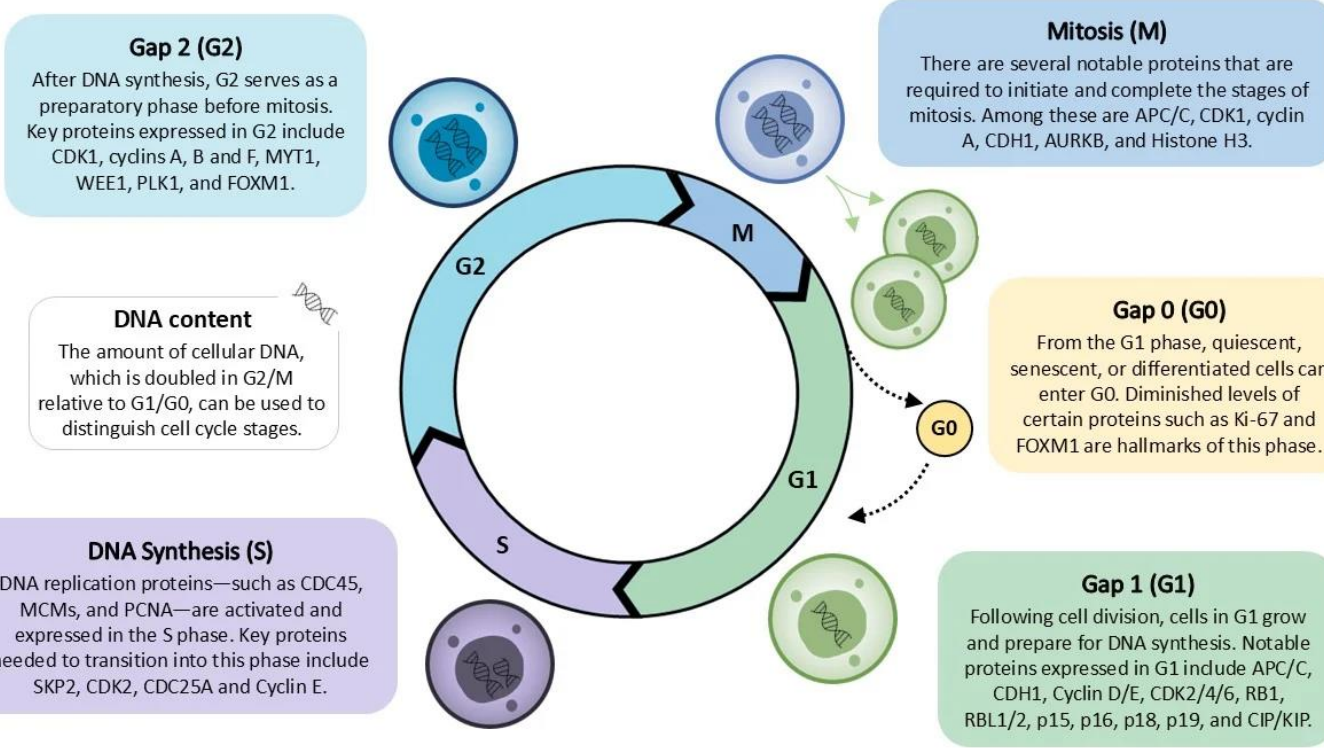


Ting Cai

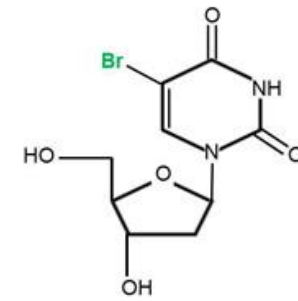
p53 present and localized normally, however, p53 is defective for transactivation



# Cell cycle

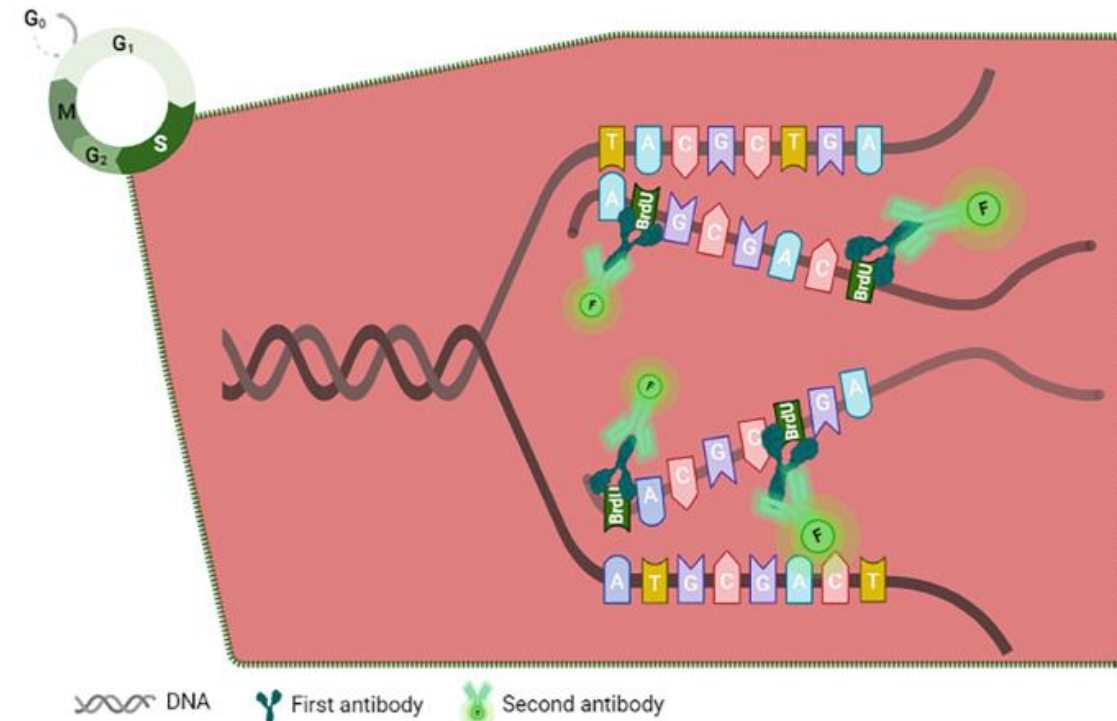


Thymidine

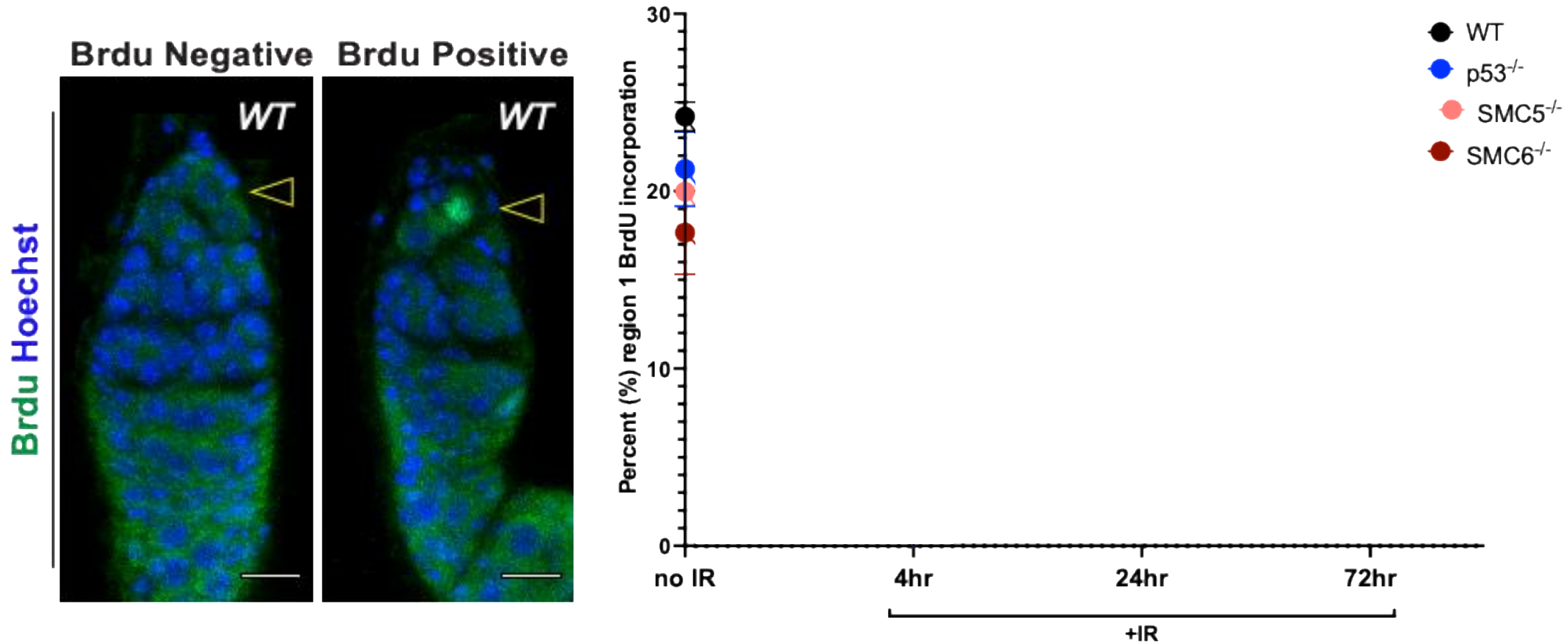


5'-bromo-2'-deoxyuridine (BrdU)

<https://static.bio-rad-antibodies.com/blog/post-images/thymidine-vs-brdu.jpg>



# Are cell cycle kinetics impaired?



p53, SMC5, and SMC6 mutants display delayed re-entry into the cell cycle after irradiation

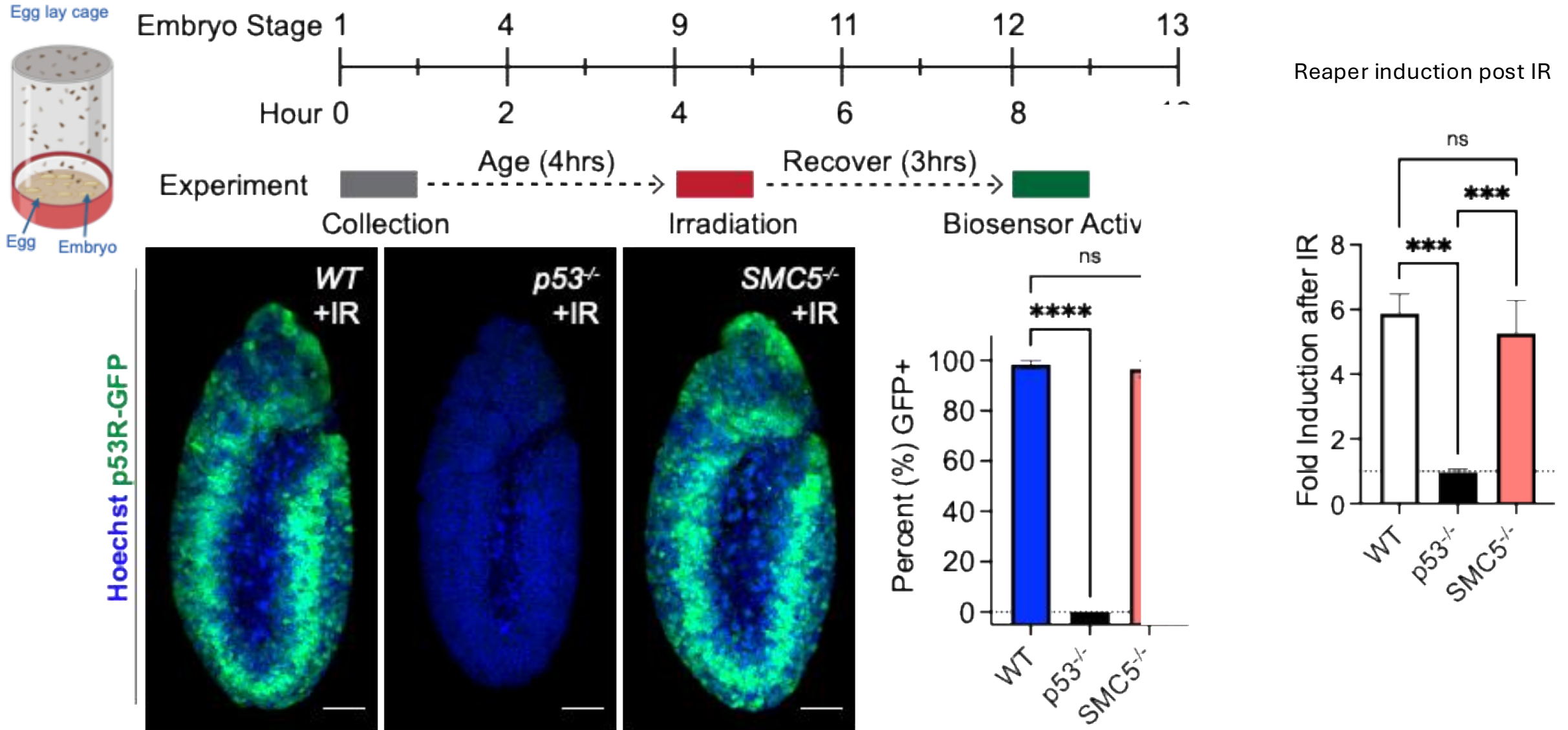
Napier-Jameson R, Cai T, Johnson A, Wylie A. The SMC5/6 complex specifies p53 activity in the *Drosophila* germline stem compartment. *Under revision for submission.*

- SMC5/6 is required for p53 activity in GSCs
- Is SMC5/6 required for p53 activity in the soma?

# Profiling the SMC5/6 complex in the soma



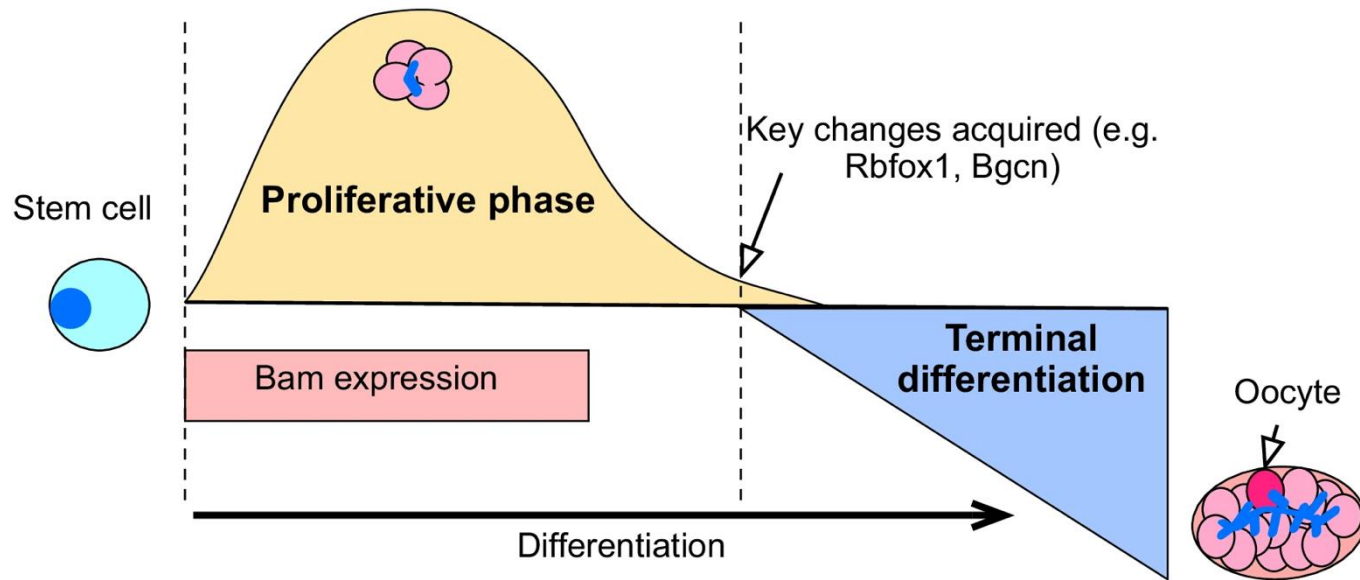
SMU | DEDMAN COLLEGE  
OF HUMANITIES & SCIENCES



*SMC5/6 complex is dispensable for p53 activity in the irradiated soma*

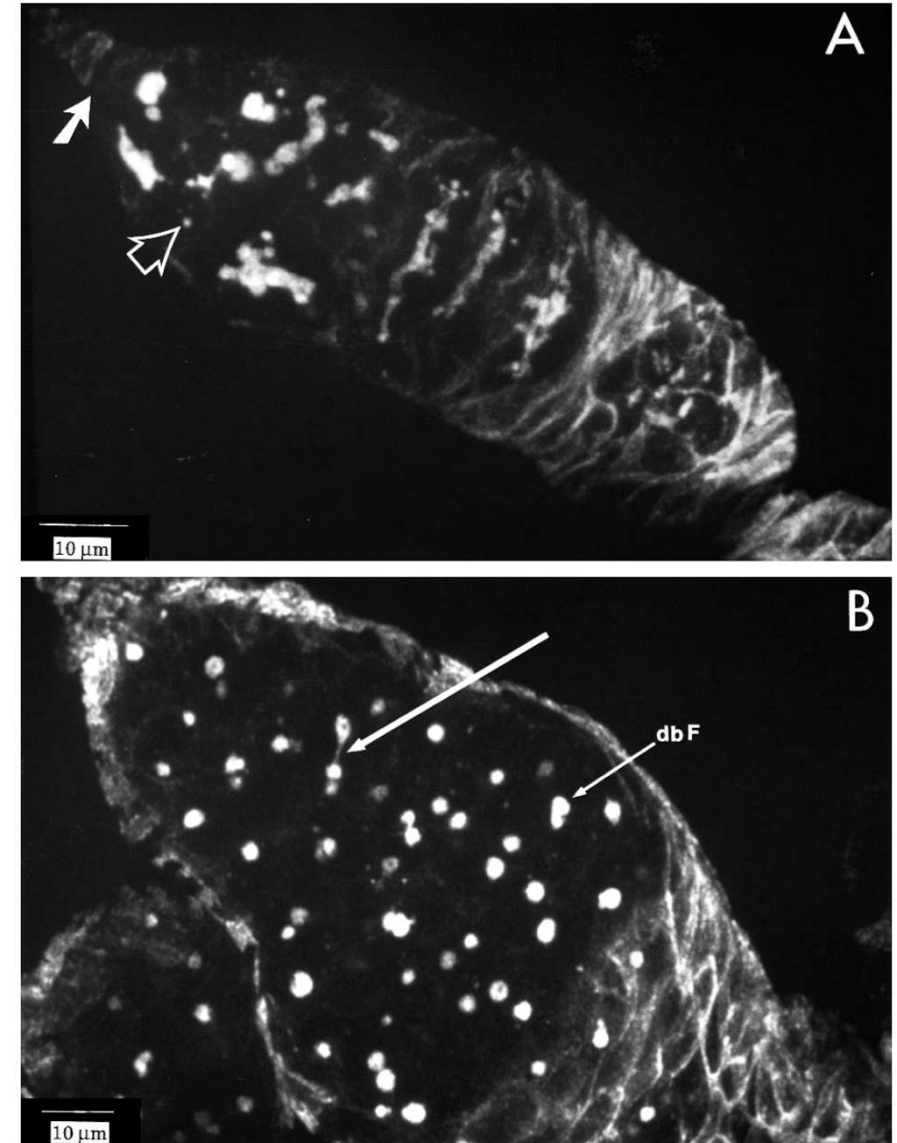


# How can we enrich in GSCs?



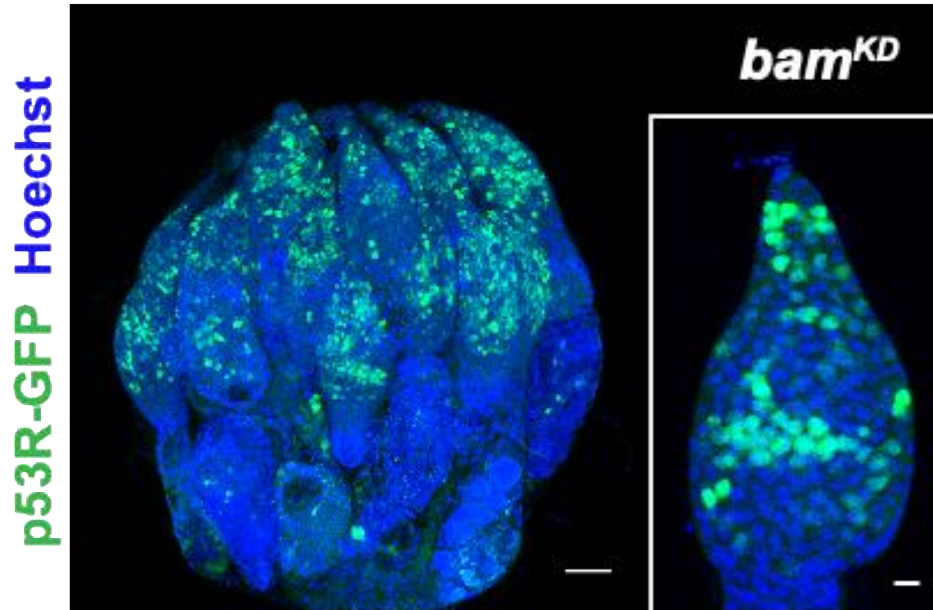
<https://doi.org/10.1038/s44318-024-00070-z>

- If bam is mutated a failure to differentiate/over proliferative phenotype occurs
- This is classified as an “oncogenic like” stress



Dennis McKearin, Benjamin Ohlstein, A role for the Drosophila Bag-of-marbles protein in the differentiation of cystoblasts from germline stem cells, *Development*, 1995, Fig. 5.

# Is SMC5/6 required for p53 activity triggered by deregulated growth in the stem cell compartment?

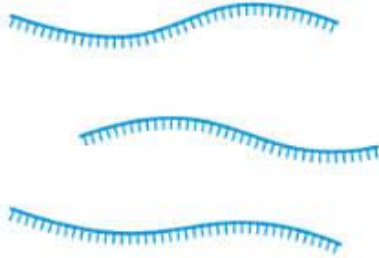


The SMC5/6 complex is required for p53 response to uncontrolled stem cell proliferation

# What is RNA-seq?

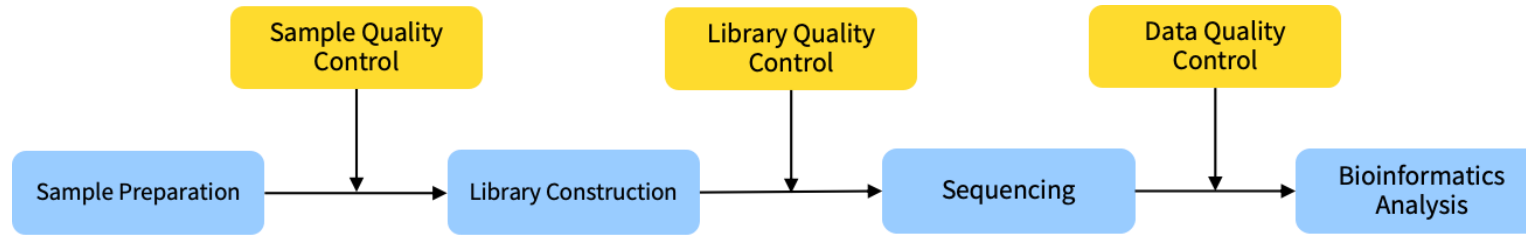
## RNA Sequencing

① Isolate RNA from samples

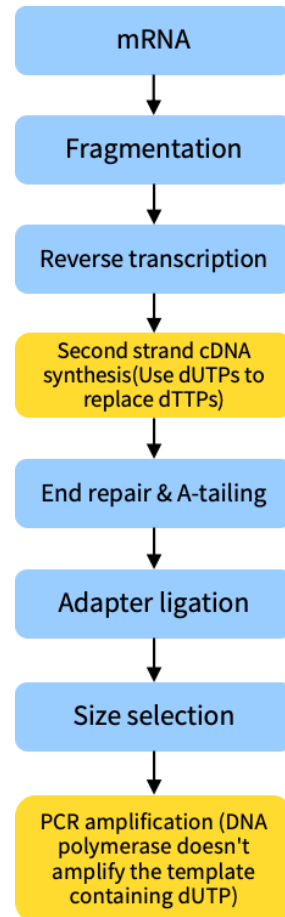


# What is RNA-seq?

## Sequence pipeline



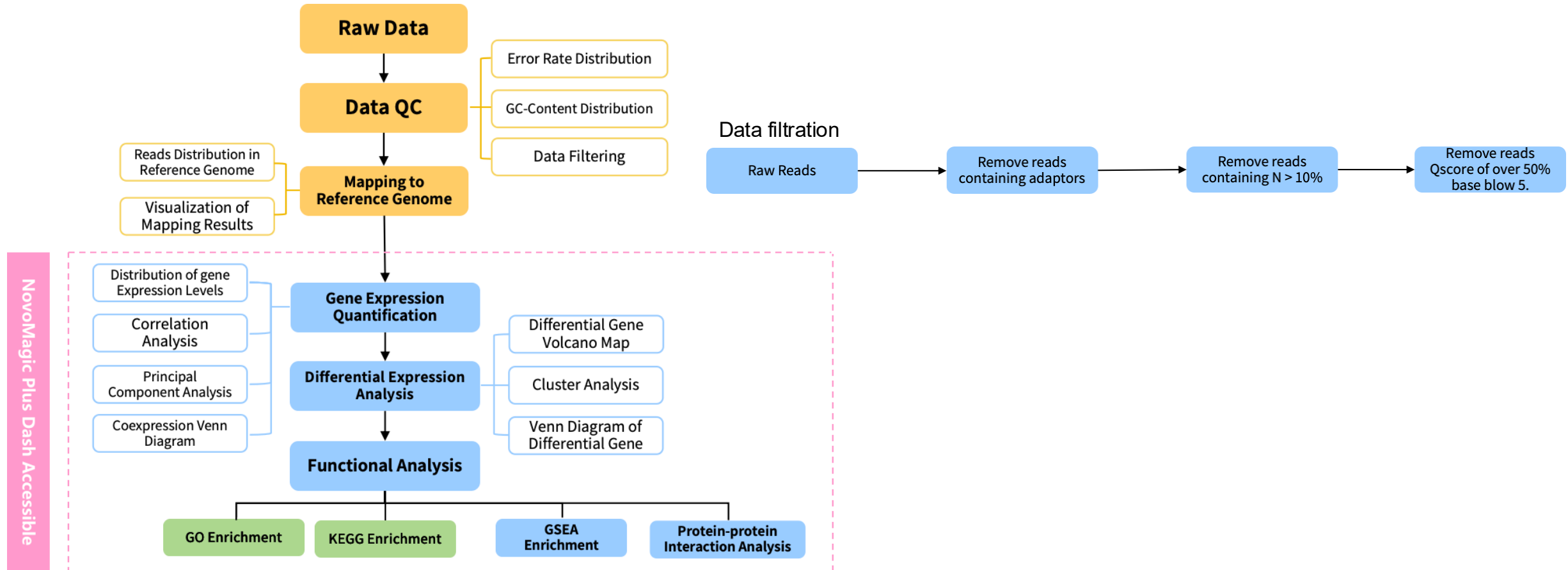
## Directional library construction pipeline



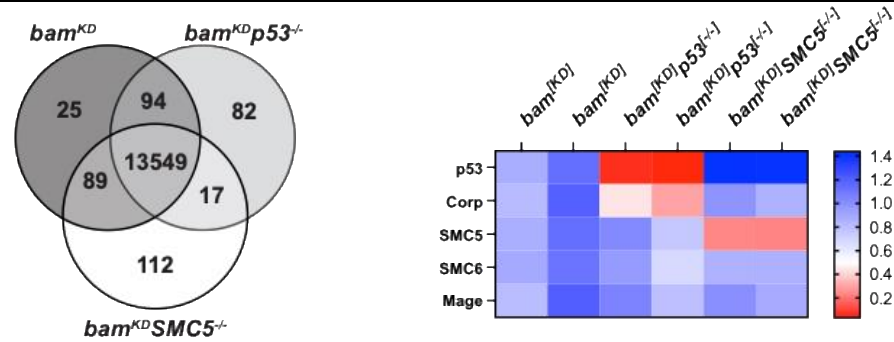


# What is RNA-seq?

## Data generation pipeline



# What transcriptomic changes do we see when enriching in GSCs?



We see many  
transcriptomic changes  
(but nothing that really stands out)

- Our biosensor provided a unique opportunity to explore tissues specific p53 activity
- SMC5/6 requisite factor for p53 activity in the germline stem cell compartment
- SMC5/6 is dispensable for p53 activity in somatic cells

Why is the SMC5/6 complex preferentially required for p53 activity in GSCs?

- Hopefully gain more mechanistic insights in terms on p53/SMC5/6:
  - In what stressor and tissue specific circumstances is each component required for p53 activity?
  - What role(s) may the complex enabled PTMs play in p53's activity/specificity?
  - Can we find more meaning in our RNA-seq data?
  - Does SMC5/6 effect p53 occupancy?
  - Is p53 part of the SMC5/6 complex?
- SMC5/6 has recently been implicated in cancer... can our studies help illuminate p53 & SMC5/6 biology that is important for tumor suppression?
- Are there other factors that specify p53 tissue specific activity?
- p53 triple tag project (still ongoing)

Hopefully, we can elucidate how SMC5/6 facilitates tissue  
(and stress) specific p53 responses

p53 is the cell's master guardian against DNA damage and cancer, understanding how it works and fails, provides critical insights into tumor development, aging, and potential therapeutic strategies



# Thanks and questions

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## People

### Wylie Lab

- **Annika Wylie**
- **Ting Cai**
- Jesiska Lowe
- Justin Wright
- Chelsea Barns
  - **Ainsley Johnson**
  - Elaine Aredo
    - Jerry Wang
    - Alex Wang

## Collaborators

### Abrams Lab (UTSW)

- Amanda Jones
- Dan Fan

## Funding



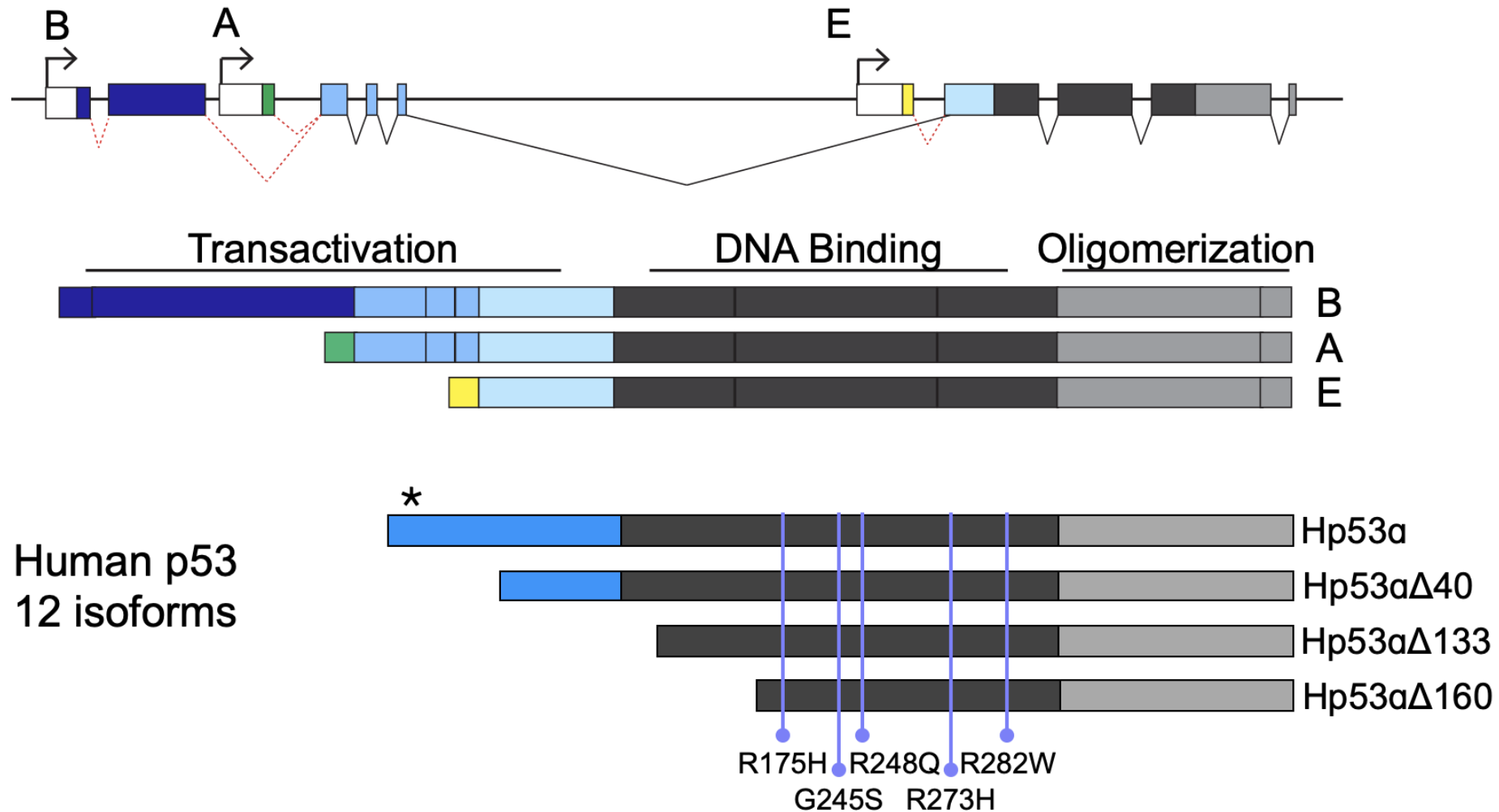
CANCER PREVENTION & RESEARCH  
INSTITUTE OF TEXAS



**A special thanks to Dr. Heidi Wu for the invitation to talk at today's workshop**

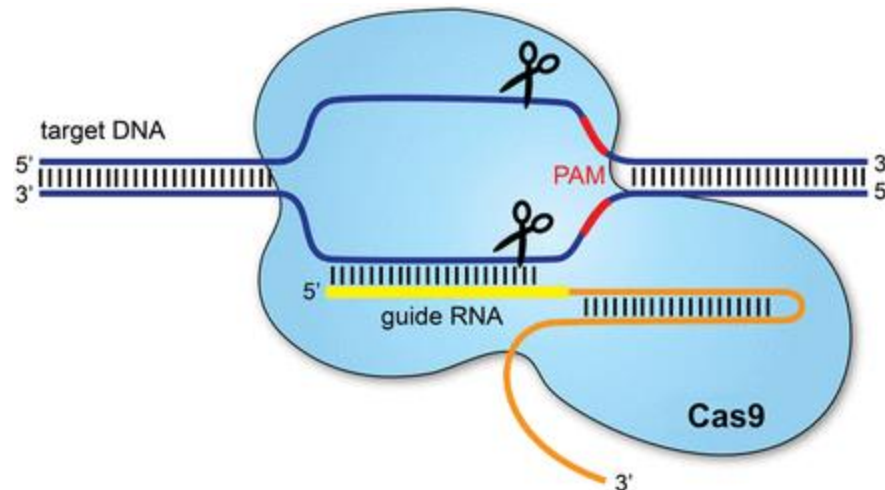
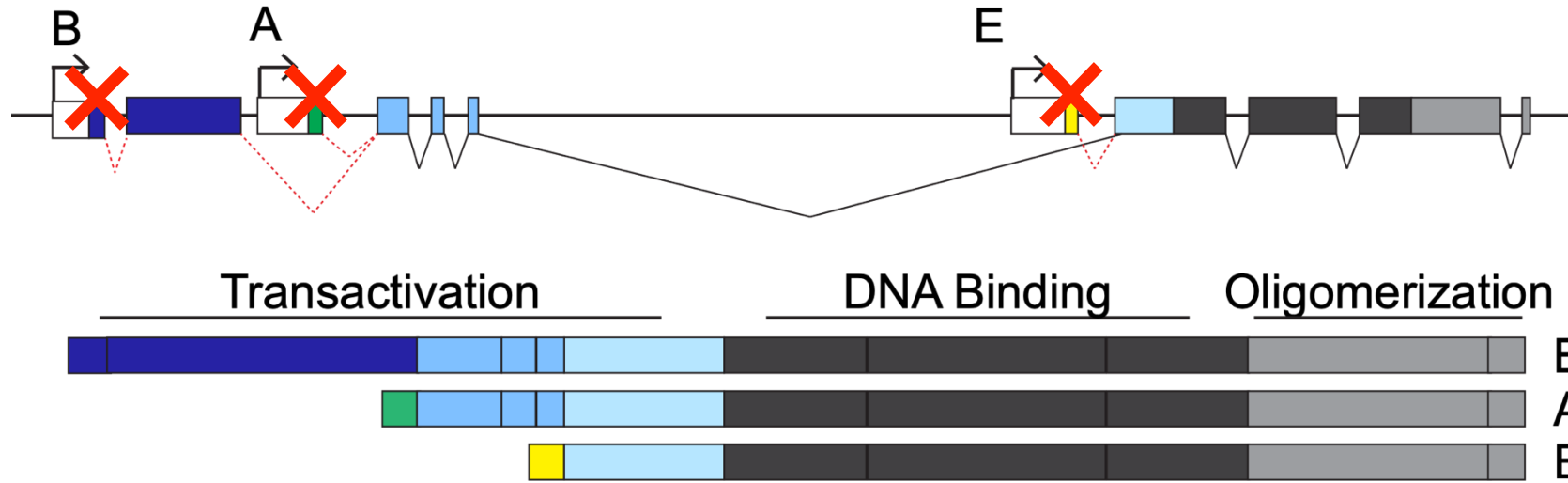


# Could isoforms account for some of the tissue specificity?



Human p53  
12 isoforms

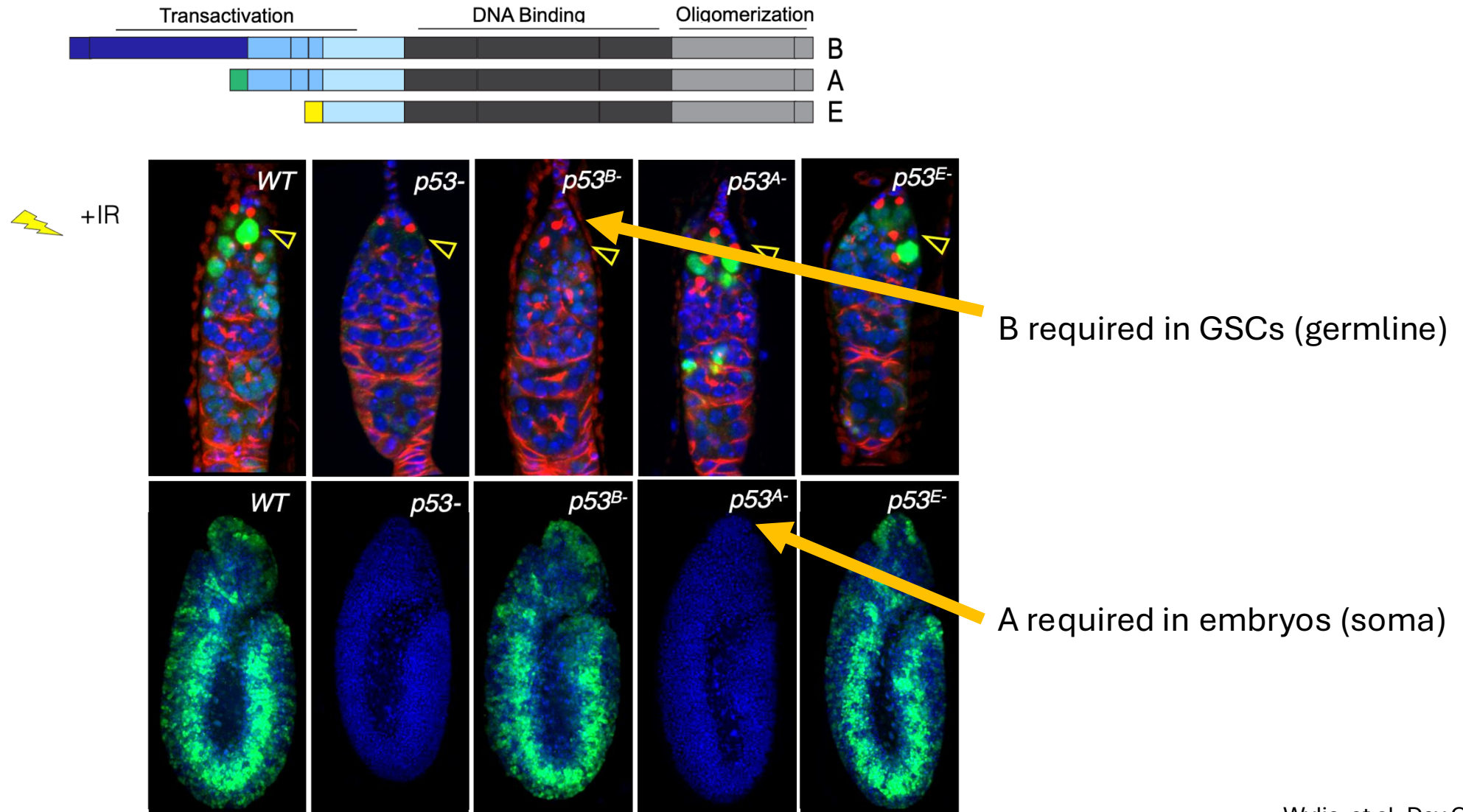
# Generating isoform specific p53 loss of function alleles



Wylie, et al. Dev Cell 2022



# Profiling p53 isoforms function in tissue-specific transactivation

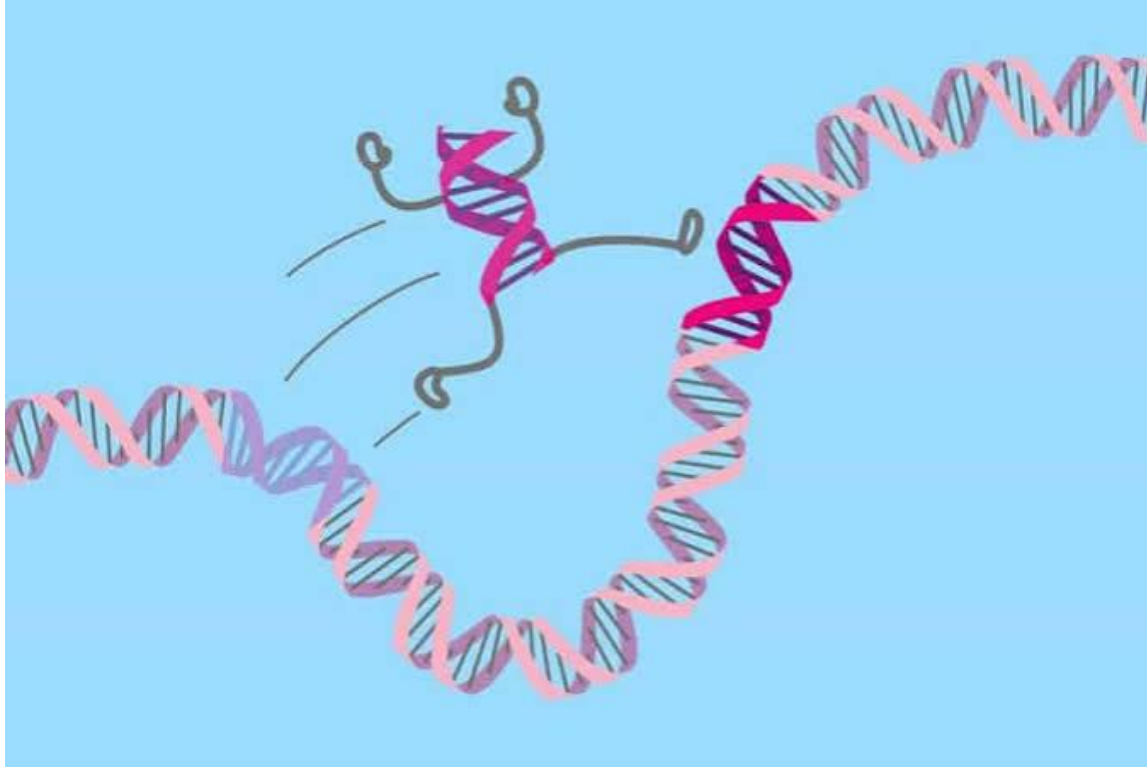


Wylie, et al. Dev Cell 2022

Could SMC5/6 be a p53 isoform specific modulator?

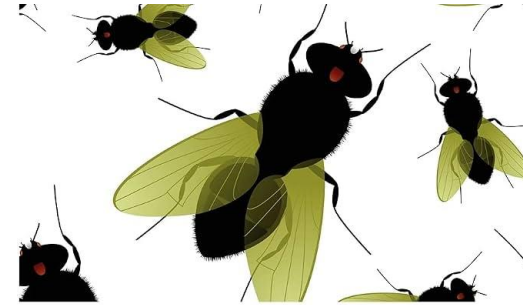
# cuff/piRNA/transposons explained...

## Jumping genes (Transposons)



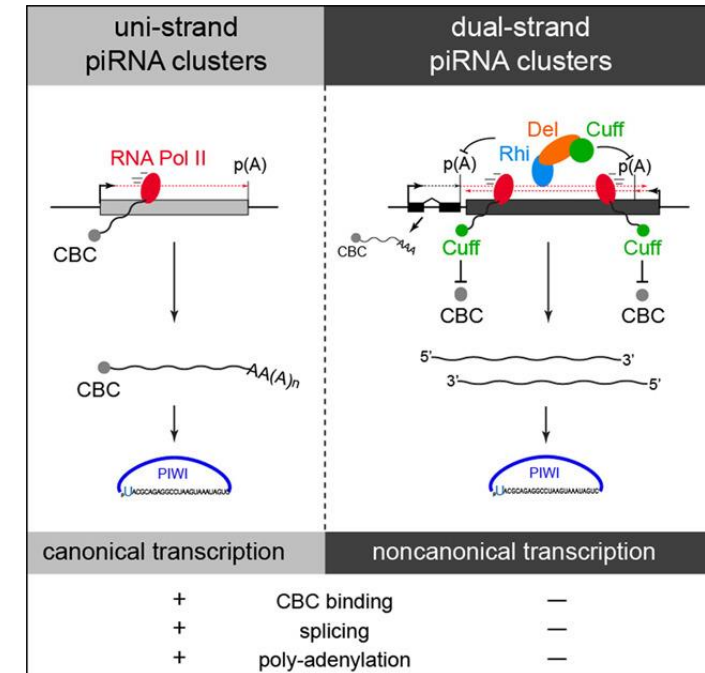
Credit: Memorial Sloan Kettering Cancer Center.

If cutoff (cuff) is mutated, the piRNA system is disrupted and a hyperactive retrotransposon phenotype is generated



Yanling Zhao  
Trudi Schupbach

### THE ROLE OF CUTOFF IN PIRNA BIOGENESIS AND TRANSPOSON SILENCING IN THE DROSOPHILA GERMLINE

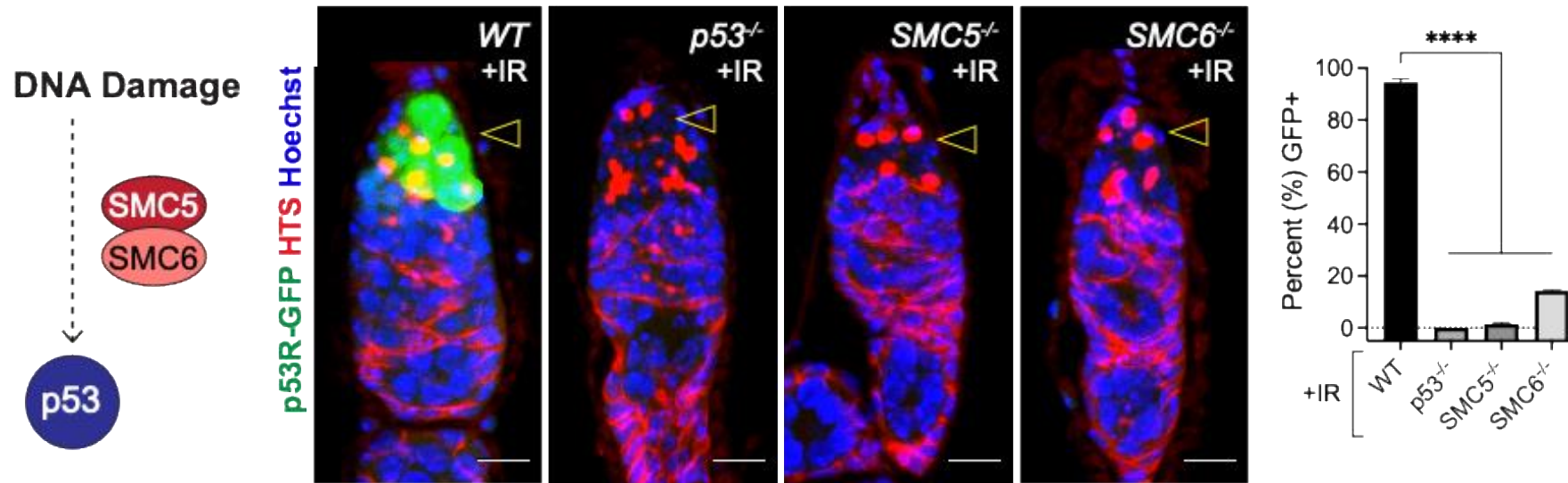


•ISBN-13 : 978-3639346220



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# But the plot thickens...



Chelsea Barnes

